

NameRank: Measuring LLM-Mediated Recognition as the Post-Bibliometric Impact Channel

Bojie Li
Pine AI

Noah Shi
University of Washington

Abstract

What a frontier model says about a person or tool is often the first description a human sees, making corpus presence a measurement problem. Citations explain about a third of whether a model recognizes a researcher; we target the residual and build **NameRank**, a $[0, 1]$ recognition score: each of 4,685 entities in 54 cohorts is probed with one open-ended question across 36 models, and an independent judge returns a binary verdict against a curated gold—did the model state a specific, non-guessable fact about this exact entity?—so hallucination, context echo, and guesses earn nothing. Synthetic-null entities hold the floor near zero, and verdicts track the entity, not the model. One thesis organizes the findings: *recognition is paid to named, indexable artifacts, not to credentials or titles*. Every Olympic-style credential sits below a working-researcher baseline, because no named artifact ships with the medal, yet the ranking inverts at the marquee tier, where Nobel, Turing, and Fields laureates saturate the panel. For independent creators the tool out-ranks its maker, and the credential that *does* propagate is a named method or awarded paper. No bibliometric predicts recognition well; top-density institutions out-recognize peers at matched citations; and on 258 news events recognition loads on peak salience, not persistence. A self-report probe shows introspection reads a corpus prior, not its own knowledge. The score stays silent rather than misleading, and the findings survive paraphrase, vintage, judge, and confound controls. We release all probes, golds, responses, and code.

Code: <https://github.com/19PINE-AI/namerank>

Website: <https://01.me/research/namerank>

1 Introduction

Ask a frontier language model about an International Mathematical Olympiad gold medalist and it will usually answer “unknown”—only about one model in eight recognizes them—while a popular open-source tool is described in detail by nearly every model on a 36-model panel. A credential that selects a few hundred people a year propagates a name less reliably than a well-named side project. This is the channel through which human curiosity about people and their work is increasingly answered: a practitioner deciding whether to read a paper, hire a candidate, attend a talk, or invest in a startup now often arrives at the question with whatever a frontier model says, before any human source intervenes. Conditional on that lookup—whose prevalence we take as a premise, not a result—what the model says is determined by whether the entity’s name has propagated into its training corpus: a clean attribution chain yields a coherent summary, its absence an “unknown” or a fabricated biography.

The information surface that mattered ten years ago—search rankings, conference plenaries, citation counts—has been re-mediated by a small number of frontier models, and whether a name has propagated into those models’ weights is now itself a substantive measurement problem.

Bibliometrics do not solve this measurement problem. In earlier work on the Incompressible Knowledge Probes (IKP) benchmark, Li [2026] found—as a byproduct of calibrating the knowledge capacity of frontier models—that citation counts explain only about a third of the variance in whether a model recognizes a researcher. A controlled audit attributed the remainder to three corpus-level mechanisms: *name uniqueness*, *named-artifact attachment*, and *subfield-ecosystem density*. That work characterized the residual but did not measure it: its recognition scores live on a coarse tier ladder built for parameter estimation, not for comparing entities. We take the opposite stance—the residual is the quantity of interest—and build a dedicated instrument for it. Section 9 details how the instrument departs from its predecessor.

NameRank measures *per-query presence*: conditional on a human asking a model about an entity, whether the model actually recognizes it. It is deliberately not weighted by how *often* an entity is queried—that human-facing exposure dimension is real but outside our scope and we do not measure it—so the score isolates the cross-entity variation in corpus presence rather than reach. Nor do we claim it measures *quality* or *merit*; it measures whether the corpora that mediate human curiosity at scale have internalized the entity’s name.

NameRank is a $[0, 1]$ recognition score. Each entity is probed with one fixed open-ended question (“tell me what you know about [name], who [context]”) across a panel of 36 frontier models. An independent LLM judge reads every response against a curated gold answer and returns a binary recognition verdict—true only when the response states a specific, non-guessable fact that is correct and *not* derivable from the disambiguating context, so a fluent hallucination, a restatement of the context, or a lucky guess contributes nothing. The per-entity score is the panel recognition rate: the probability that a randomly drawn frontier model genuinely knows the entity. Three properties distinguish this from recognition numbers that fall out of benchmark suites: it places researchers, founders, open-source maintainers, and named artifacts on a single axis; it is hallucination-resistant by construction; and cross-model disagreement is retained as signal rather than averaged away.

A single thesis runs through the results and organizes the paper. *Recognition in the LLM channel is paid to named, indexable artifacts—not to credentials, titles, or reach*. A credential propagates a name only to the extent that named production attaches to it, and one axis makes this legible: an olympiad medal, which ships with no named artifact, sits near the floor; a best-paper award, which ships with a named paper, sits well above the working-researcher baseline; authorship of a widely-used named method sits higher still; and the marquee prizes, backed by a career of named production, sit at the top against the panel ceiling. The artifact-over-creator inversion—where a tool out-ranks its maker and no prompt can manufacture the reverse—is the mechanism seen most directly; the institution gradient and the news-event calibration are the same mechanism from other angles.

Contributions.

1. A **cross-domain recognition instrument**—a binary recognition verdict over a 36-model panel, calibrated by synthetic-null floors—that places researchers, founders, OSS authors, public websites, and named artifacts on one $[0, 1]$ axis (Section 2, 3).

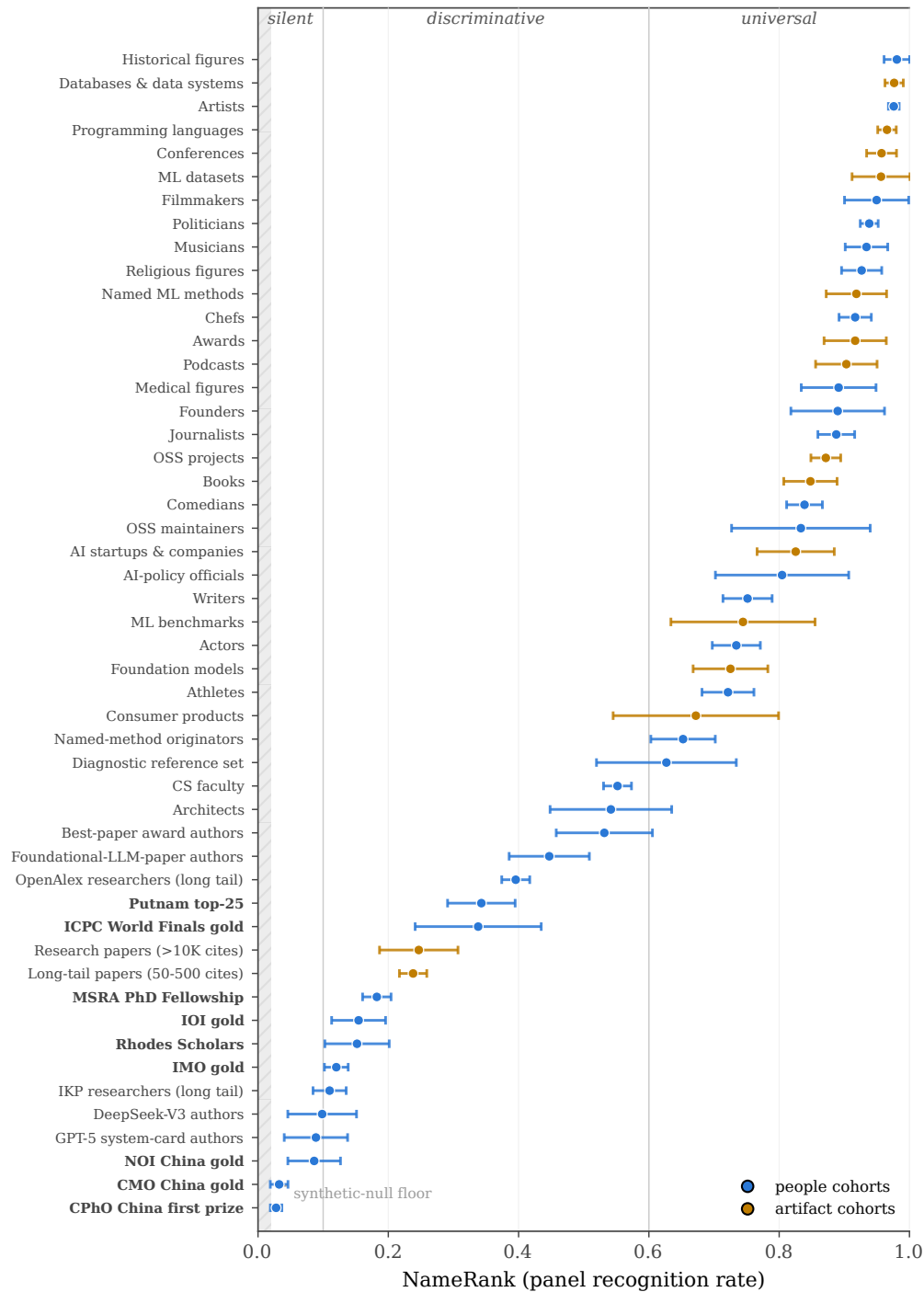


Figure 1. The recognition atlas. Every cohort ($n \geq 10$) placed on the single NameRank axis: dot is the cohort’s mean panel recognition rate, whisker the 95% CI over entities; people cohorts in blue, artifact cohorts in amber. The hatched band is the synthetic-null floor (fictional entities of the same recipe). Three zones emerge on one scale: *silent* (at floor), *discriminative*, and *universal*. The nine olympiad/fellowship credentials (bold) all fall at or below the working-researcher cohort, while named artifacts dominate the universal zone—the paper’s thesis in one exhibit (Section 3).

2. The **career-arc account of credentials**: every Olympic-style credential scores at or below a working-researcher baseline, while best-paper awards, named-method authorship, and marquee prizes rise above it—a unified picture in which a credential propagates a name in proportion to the named artifacts it carries (Sections 4–4.3).
3. The **named-artifact mechanism**: for independent creators the artifact out-ranks its maker; the effect is an observational property of corpus attribution, not something a context intervention can manufacture, and it dissolves the tempting reading of recognition as reach (Section 5).
4. **What predicts recognition**: no bibliometric stands in for recognition— h -index, citations, and attribution-weighted variants all explain only about a fifth of the variance; a corpus-density gradient runs across institutions (even at matched citations) and countries; and a 258-event news cohort with a recorded attention ledger calibrates the instrument externally—peak salience, not persistence (Sections 6–6.3).
5. A **validation battery** over wording, vintage, judge family, synthetic nulls, and gold confounds, plus a **self-report probe** showing introspection reads the corpus prior rather than the model’s own state (Sections 7–7.1); and an **open release** of all probes, golds, responses, and code.

Figure 1 previews the central result: a single recognition axis on which every cohort is placed.

2 Measuring NameRank

Figure 2 summarizes the pipeline this section unpacks: a fixed open-ended probe is sent to a 36-model panel for each entity; an independent LLM judge reads each (entity, model) response against a curated gold answer and returns a binary *recognition* verdict—did the response demonstrate genuine memory of this specific entity; and the per-entity NameRank is the fraction of the panel that recognizes it.

2.1 Definition

For an entity e with a disambiguating context $c(e)$ and a curated gold answer $g(e)$, and a model panel $\mathcal{M}$ of size $M = 36$, NameRank is the fraction of the panel that recognizes the entity,

$$\text{NameRank}(e) = \frac{1}{M} \sum_{m \in \mathcal{M}} \text{rec}(r_{e,m}, c(e), g(e)), \quad (1)$$

where $r_{e,m}$ is the response of model m to the open-ended probe

“Tell me what you know about [name], who [context]. If you do not recognize this entity, respond with ‘unknown’. Limit your response to about 150 words.”

and $\text{rec} \in \{0, 1\}$ is a recognition verdict produced by an LLM judge (Section 2.4): it is 1 only when the response states at least one *specific, non-guessable* fact that is true of this exact entity—verified against the gold answer or the judge’s own knowledge—and not derivable from the disambiguating context. A refusal, a hedged non-answer, a response confined to context-derivable generalities, and a fluent but wrong-person biography all score 0. NameRank thus

NameRank pipeline. Open-ended probe → 36-model panel → binary recognition verdict per record → panel recognition rate.

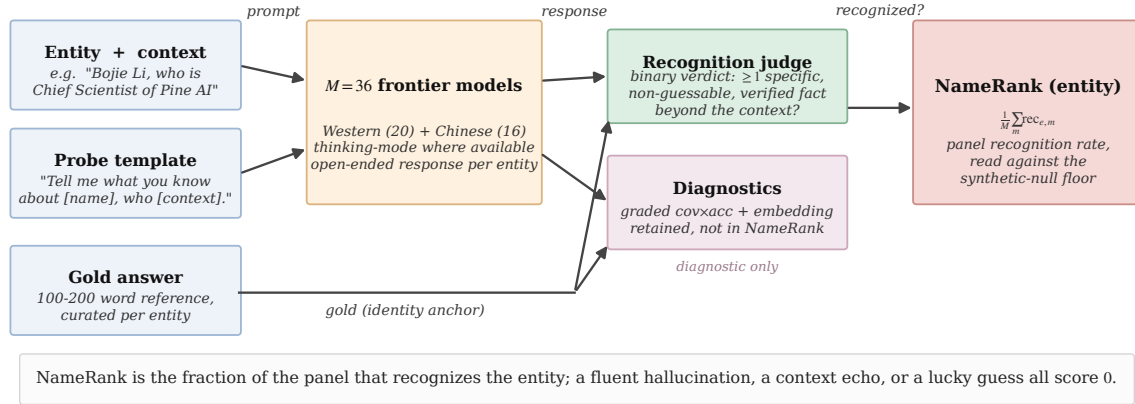


Figure 2. NameRank pipeline. The probe template is fixed, with two slots: the entity name and a short role/affiliation/field context. Each of the 36 panel models answers the open-ended probe; an LLM judge reads the response against the gold answer and returns a recognition verdict—true only if the response states at least one specific, non-guessable fact about the entity that is correct and not derivable from the context. The per-entity score is the panel recognition rate. An embedding similarity (BAAI/bge-large-en-v1.5) and a graded coverage × accuracy score are stored per record as diagnostics only—neither enters the headline score (Appendix J). Totals: 234,574 recognition verdicts over 6,512 entities (54 core cohorts plus award, event, LLM-area, and university extensions), plus a 240-entity Chinese-prompt sub-run.

reads as the probability that a randomly drawn frontier model genuinely knows the entity—concretely, a NameRank of 0.12 means about four of the 36 models recognized the entity while the rest refused or hallucinated, and 0.90 means essentially all did. Three worked (entity, model, response) triples spanning the range are provided in the [repository documentation](#).

2.2 Probe Design

One open-ended probe per entity. Closed-factoid probes have a known failure mode: a model that recognizes the entity through fact X but does not recall fact Y scores zero on a probe-of-Y, biasing the metric toward specific-fact recall. The open-ended probe instead asks the holistic question: does the model have a retrievable representation of this entity? The template is fixed, with two slots: the canonical entity name, and a short disambiguating clause giving role, affiliation, and field (e.g., “Wei Zhang, who is a professor of mathematics at MIT working on PDEs”). The context resolves name collisions at probe time without revealing the gold answer. The full English and Chinese probe text and the judge prompt are reproduced verbatim in the [repository documentation](#).

The context names role, affiliation, and field—never specific contributions, papers, dates, or named artifacts—so it anchors disambiguation without leaking the gold; the judge’s rubric credits only specific named facts, and generic in-context biographies earn nothing (evidence in Appendix A; a dedicated context-ablation in Appendix O).

2.3 Gold Answers

For each entity we construct a gold answer built from *entity-specific, verifiable facts*—named works, named contributions, real affiliations with dates, and other identifying particulars—composed from primary and authoritative sources: bibliographic records (OpenAlex, Semantic Scholar, Crossref) for researchers and papers, official competition and award records for credential holders, code-host metadata and READMEs for open-source artifacts, encyclopedia lead sections for public figures and companies, and length-normalized web-grounded profiles for the long tail. The gold serves two roles for the judge: it *identifies* which specific entity the probe refers to (disambiguating same-name individuals), and it supplies a correctness anchor. It deliberately does *not* restate the disambiguating context: a fact the probe already supplies earns no recognition credit, so the gold’s discriminating content is the material a model must know independently. We audit a stratified sample against primary sources following the protocol of Li [2026]; the observed correction rate is small, typically year-of-affiliation drift. A confound check confirms NameRank does not reduce to a has-a-Wikipedia-page flag (Appendix O).

2.4 The Judge

We use Gemini 3 Flash Preview as the judge (direct Google API to minimize routing variance; temperature 0). The judge is shown the entity name, the probe context, the gold answer, and the model’s response, and returns a binary *recognized* verdict. It is instructed to credit recognition only when the response states a specific, non-guessable fact that is (a) true of this exact entity, established either by the gold answer or by the judge’s own reliable knowledge; (b) not derivable from the probe context; and (c) about the same entity the gold identifies, not a same-name individual. The judge uses knowledge *asymmetrically*: it may refute freely—flagging a wrong-person biography or a false claim using anything it knows—but it may *credit* a fact only when it positively verifies it, so a plausible-sounding but unverified specific about an entity it does not recognize counts for nothing (the signature of hallucination on obscure or fictional names). For competition participants, fellowship recipients, and minor-paper authors—where an LLM judge has no reliable per-individual memory—an anti-confabulation provision restricts credit to facts present in the gold or to major, widely-documented achievements, never to fine-grained particulars asserted from the judge’s own memory. Broad field, nationality, an employer inferable from the name, generic praise, and restatements of the context are all non-credit by construction.

We also record, per response, a graded coverage $\times$ accuracy score and the embedding cosine similarity to the gold [Xiao et al., 2024]; both are retained as diagnostics and neither enters the headline NameRank. Appendix J shows why a judge is required—embedding similarity credits fluent hallucinations as recognized—and why the recognition verdict is preferred to a graded score: graded coverage measures overlap between two arbitrary samples of an entity’s fact universe, penalizing a model that knows the entity through facts the gold happens to omit, whereas recognition asks the construct-faithful question of whether the model knows the entity at all. The dependence of the findings on the judge’s model family is quantified with a multi-judge re-grade in Appendix M.

2.5 Aggregation

NameRank is the unweighted panel recognition rate across the 36-model panel (Equation 1). We also retain, per entity, the refusal rate, the Western/Chinese sub-panel rates, the diagnostic graded score, and the raw response text. Averaging across the panel is what makes the score stable: a single model contributes noise from prompt sensitivity, refusal-policy idiosyncrasies, and training-data accidents, while the panel rate integrates corpus presence over the current frontier fleet. The retained per-model verdicts preserve the residual that the panel mean averages over—per-model generosity and vendor-family differences, tabulated in the [released per-model statistics](#).

Calibration. Two properties make the resulting axis trustworthy (Figure 3). First, a *synthetic-null floor*: for each cohort we construct fictional entities on the same gold recipe—plausible but non-existent researchers, founders, medalists—and run them through the full pipeline. Because the judge credits only positively-verified facts, these earn essentially zero (≤ 0.02 for people and artifacts), so a nonzero score is real recognition rather than a guessing artifact; the one exception is fictional papers, whose descriptive titles admit plausible guesses (0.06), and this floor is subtracted where relevant. Second, a *variance decomposition* of the per-record verdicts: the entity factor explains 56% of the variance against 9% for the model, so the score measures the entity, not the panel. Scores are read against the floor throughout (Appendix N, J).

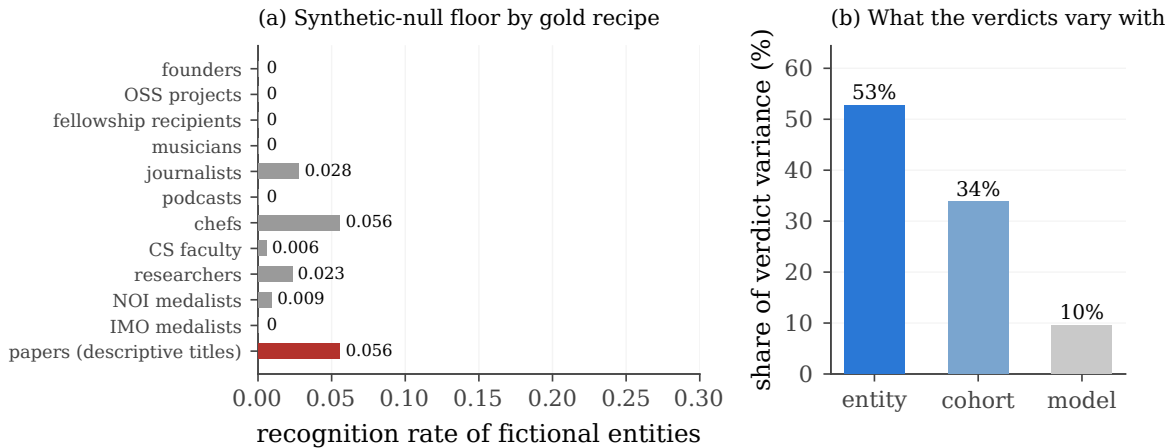


Figure 3. The instrument is calibrated. (a) Fictional entities of each gold recipe earn near-zero recognition—the guessing floor—except descriptive paper titles (subtracted where relevant). (b) Per-record verdicts vary overwhelmingly with the entity, not the model.

2.6 Panel, Cohorts, and Release

The panel comprises 36 frontier and near-frontier models accessed via OpenRouter and direct vendor APIs, spanning Western and Chinese vendors and frontier-reasoning to small (sub-30B) classes, with thinking mode enabled wherever a variant exists—the operational mode in which most user queries are answered (full roster in Appendix A, Table 2). The headline NameRank is the unweighted mean over all 36 models; the Western/Chinese partition enters only in the cross-language robustness check (Section 7.2), where both model classes are shown to shift together on Chinese prompts.

We evaluate 4,685 entities in 54 core cohorts—credentialed individuals (olympiad medalists, Putnam, ICPC, Rhodes, MSRA fellows), working academics, industry figures, named artifacts, and mid-tier cross-profession figures (writers, athletes, actors, chefs, podcasts)—plus a hand-curated diagnostic reference set and special cohorts (technical-report and system-card author lists, conferences, awards); award, event, LLM-area, and university extensions are probed under the same protocol (per-group breakdown in Appendix A, Table 3). The design goal is coverage of the axes the metric claims to unify: credentialed individuals, whose recognition should be carried by the credential *if* credentials propagate; working academics with known bibliometrics, as the external-validity anchor; industry figures whom no bibliometric measures; named artifacts, to compare things against their makers; and mid-tier figures, to test the axis outside the technology world. A 240-entity subset was re-probed with a Chinese-translated template (Section 7.2). We release the probe specifications, all gold answers, all judged response records, the Chinese sub-run, and the analysis pipeline at <https://github.com/19PINE-AI/namerank> (companion website: <https://01.me/research/namerank>).

3 The Recognition Atlas

Figure 1 places every cohort on the single recognition axis, and three properties of the instrument are visible at once. First, the axis is *meaningful across domains*: researchers, founders, open-source tools, competition medalists, and named methods all land on one $[0, 1]$ scale, something no bibliometric or domain-specific metric supports. Second, the score is *silent, not misleading, below threshold*: the lowest cohorts sit inside the hatched synthetic-null band, where the panel withholds recognition rather than fabricating rankings. Third, an *artifact/person split* is already legible—amber artifact cohorts crowd the universal end while the bold olympiad credentials sit at the bottom.

Three zones organize the axis. The *silent* zone (at the floor) holds names that have not propagated: GPT-5 system-card authors (0.09) and DeepSeek-V3 authors (0.10), each a list of real, accomplished engineers whose names appear only inside a large contributor roster. The *discriminative* zone is where the instrument does most of its work: working researchers (0.40), CS faculty (0.55), and the credential cohorts spread across it. The *universal* zone holds named artifacts and public figures—named ML methods (0.92), OSS projects (0.87), programming languages and AI companies near the ceiling—that essentially every model knows.

Figure 4 makes the cross-domain claim concrete by naming individual entities. A best-selling framework, a Fortune-10 CTO, an OSS maintainer, and a viral website land on the same axis that carries Yann LeCun (1.00) and Geoffrey Hinton (1.00): citation count does not measure a Walmart CTO, *h*-index does not measure the creator of nanoGPT, and GitHub stars do not measure Sam Altman, but recognition measures them all. The instrument’s design—which makes this axis trustworthy—is the subject of Section 2; its calibration (Figure 3) shows fictional entities earning essentially zero and the verdicts varying with the entity, not the panel.

4 The Credential Treadmill

The atlas already shows every olympiad and fellowship credential sitting at or below the working-researcher baseline. This section reads that pattern—and its inversion at the marquee tier—as the most visible consequence of a single mechanism: recognition attaches to named,

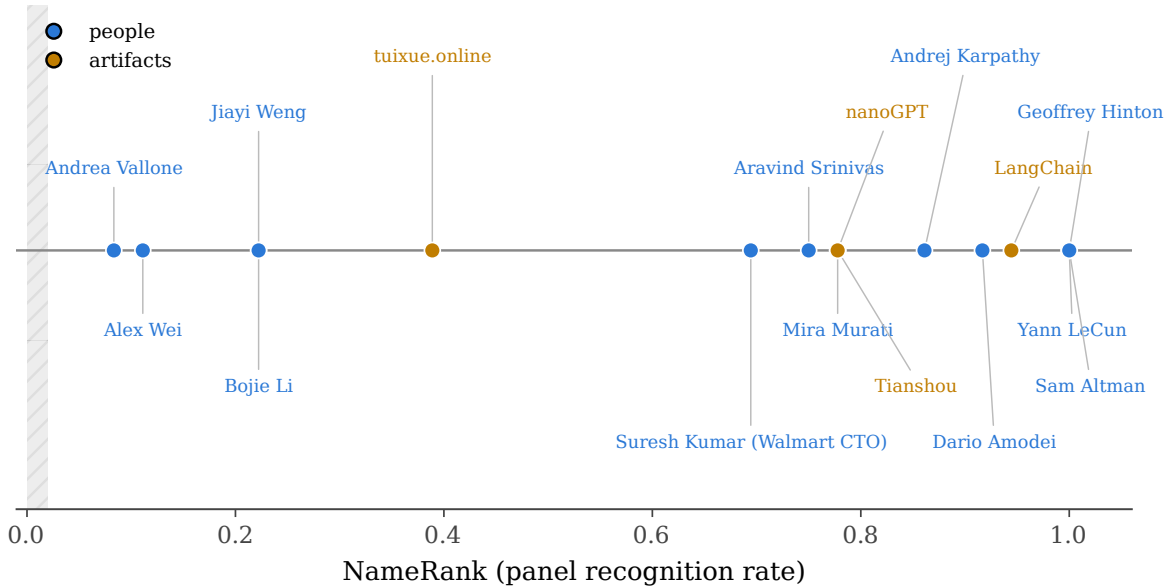


Figure 4. One axis, every domain. Selected named entities on the recognition axis; people in blue, artifacts in amber. Citation counts, h -index, and GitHub stars each measure only a slice of this range; recognition places all of them together. The hatched band is the synthetic-null floor.

indexable artifacts, not to the credential itself, so a credential propagates a name only to the extent that named production follows it. Section 5 then isolates that mechanism causally.

4.1 Nine credentials, all below the baseline

We place nine competition and fellowship credentials—each once a career-defining distinction—on the recognition axis against a baseline of ordinary citation-tracked researchers (OpenAlex, recognition 0.40; Figure 5). *Every one falls below it.* The two nearest—Putnam Fellows and ICPC World Finalists (0.34 each)—are exactly the credentials that feed almost exclusively into US-tracked, English-corpus careers, and even they fall short; the globally-dispersing olympiads sit near the synthetic-null floor (IMO 0.12, CMO and CPhO at 0.03). Below even the credentials sit two large-team author-list cohorts (DeepSeek-V3 report, GPT-5 system card): membership in a famous project, absent an individually named artifact, propagates almost nothing.

Two properties confirm that the career, not the medal, carries the signal. First, *career track dominates*: within IMO gold medalists, US-tracked academic and industry careers—the subset comparable to the citation-floored baseline—score materially higher than non-Anglophone-tracked golds, which is why Putnam and ICPC come closest to the baseline while IMO and CMO fall furthest below it. Second, *recency does not matter*: medal year is essentially uncorrelated with recognition, so what accrues is post-medal production rather than the medal’s freshness (Section 8.1). The medal is not an exception to recognition decay but a proxy for selection into careers that later produce English text—and only the production propagates the name.

The generational claim. Olympic-style credential systems were optimized for a pre-LLM signaling regime in which a distinction translated into reputation through human social networks [Spence, 1973]. In the LLM-mediated regime, only credentials that translate into *named, indexable production*—papers, tools, blog posts, talks carrying the holder’s name—

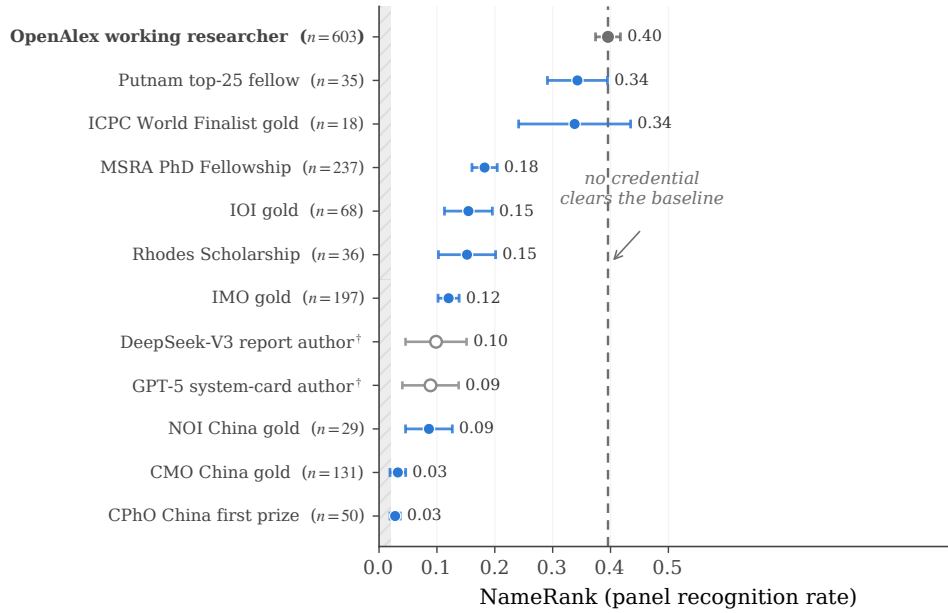


Figure 5. The credential ladder. The nine olympiad/fellowship credentials on the recognition axis (blue, 95% CI), against the working-researcher baseline (dashed, 0.40) and the synthetic-null floor (hatched, ~ 0.02). Every credential falls below the baseline; Putnam and ICPC—feeding US-tracked careers—come closest, while the globally-dispersing olympiads sit near the floor. [†] Two large-team author-list cohorts (not credentials) shown for context; their full probed rosters (69/80) exceed the entity-specific-gold subset scored here.

propagate. The medal itself, absent subsequent production, does not.

4.2 Inside one contest: what the medal actually buys

Probing a single contest’s full roster isolates the credential from the career. On the complete China National Olympiad in Informatics 2009 roster—every gold, silver, and bronze medalist—the gold/non-gold cliff survives (gold recognition 0.10 against 0.03/0.02 for silver/bronze, Kruskal–Wallis $p < 10^{-5}$, against a near-zero synthetic-null floor), while contest *score* within a tier does not predict recognition at all. The few medalists who rise above the near-floor band are exactly those whose *post-medal* careers produced named work—a now-prominent vision professor, a research lead, the author of this paper on a systems-and-startup track (Appendix C). The medal grade sets the odds of entering a documented cohort; the recognition tracks what the person later built.

4.3 The credential that does propagate: named production

If the mechanism is named-artifact attachment, then a credential that *ships with a named artifact* should propagate. The LLM-era publication credentials confirm it directly (Figure 6): authors of best-paper-awarded papers at top venues score 0.53 and named-method originators 0.65—both above the working-researcher baseline, because the award or method carries a named, heavily-indexed artifact into the corpus alongside the person, and rising in exactly that order as the artifact attaches more directly to the individual. The marquee prizes sit higher still, at the panel ceiling (Turing, Fields, Nobel laureates all above 0.95). Two reading notes temper the

marquee tier. First, at that ceiling the prize marks fame as much as it made it: an individual laureate is a famous person the judge recognizes directly, and because recognition saturates there is no headroom for a within-award time trend to read a lifetime clock off the marquee tier (that clock is visible only in the *unsaturated* cohorts; Appendix B). Second, the sampled mid-career fellowship cohorts are upper bounds—they include only recipients prominent enough to have an encyclopedia article. The clean comparison is among complete-roster cohorts on identical rosters, and along that spine the arc is monotone.

The groups of Figure 6—olympiad, LLM-era publication, mid-career honor, marquee prize—trace it: recognition rises with the density of named artifacts a credential carries, not with the intellectual selectivity of the filter that awarded it. This is the paper’s organizing result.

5 The Named-Artifact Mechanism

If recognition is paid to named artifacts, the artifact should sometimes out-rank the person who made it. It does, and the direction of the effect—plus a control that comes back null—pins the mechanism to corpus attribution rather than anything a prompt can manufacture.

5.1 The artifact-over-creator inversion

For verified (creator, artifact) pairs we measure both members independently (Figure 7a). In 7 of the 10 resolvable pairs, the artifact out-ranks its creator, and for independent creators of a single artifact the inversion is uniform: Jiayi Weng (0.22) below Tianshou (0.78), Harrison Chase below LangChain, Aman Sanger below Cursor, Simon Willison below Datasette, Tri Dao below FlashAttention, Aravind Srinivas below Perplexity. The three exceptions—Demis Hassabis, Andrej Karpathy, Mira Murati—are senior leaders who accumulated independent recognition before, or beyond, the artifact in question; Karpathy is famous for too many things for nanoGPT to carry him. The pattern extends beyond these hand-verified pairs: among named methods prominent enough to appear as standalone entities, the method out-ranks its originator in almost every case (Figure 7b), and the named-method-originator cohort as a whole (0.65) sits far above the researcher baseline precisely because each member’s method drags their name into the corpus.

The direction is an asymmetry in the corpus itself, not a symmetric association. Measuring how often a probe about one member names the other, the common regime is *famous artifact, anonymous creator*: a probe about the creator reliably surfaces the artifact, but a probe about the artifact frequently omits the creator. Open-source and academic norms produce the opposite, balanced pattern (creator and artifact co-occur in both directions), and the inverse case—Karpathy and nanoGPT—shows a probe about the artifact reliably surfacing the creator while the reverse fails (per-pair breakdowns in Appendix D). This attribution asymmetry is exactly what makes the artifact out-rank its maker: the corpus discusses the tool far more often than the person behind it.

5.2 The inversion is observational, not inducible

A natural causal question: if we *tell* the model the creator made the artifact, does recognition of the creator rise? We re-probed all 11 pairs under a controlled intervention—configuration A gives a role-only context, configuration B appends one clause naming the artifact—and scored

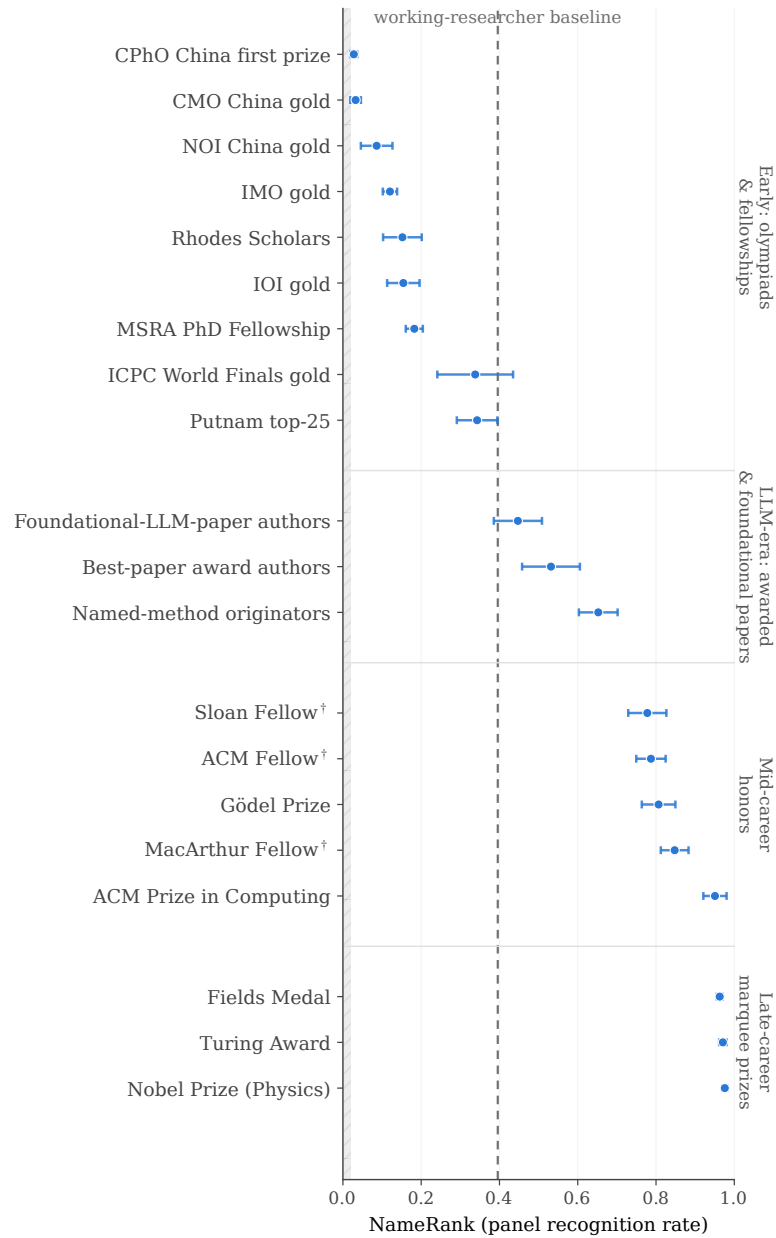


Figure 6. The career arc, end to end. All credential and award cohorts by career stage on the recognition axis, against the working-researcher baseline (dashed) and the synthetic-null floor (hatched). Early olympiad/fellowship credentials sit at or below the baseline (detailed in Figure 5); LLM-era publication credentials (best-paper authors, named-method originators) and marquee late-career prizes rise above it. Recognition tracks the named production a credential carries, not the selectivity of the award.

both with the recognition judge, which credits only facts *beyond* the context. The lift is essentially zero (mean -0.01 ; positive only for the two least-known creators, Jiayi Weng and Tri Dao). This is the informative null: an earlier scoring rule that credited coverage showed an apparent lift, but that lift was the model echoing the injected artifact name back: once the judge stops rewarding the echo, naming the artifact adds nothing. Recognition of the creator is written into the corpus or it is not; a prompt cannot manufacture it. The inversion is thus an observational property of how attribution is distributed in indexable text, which is exactly what the mechanism predicts.

5.3 Boundary condition: reach without name propagation

The mechanism also says what recognition is *not*. *tuixue.online*, a Chinese-language site that monitored US visa-appointment availability with a peak user base near one million, has larger human reach than any open-source tool in our comparison set, yet scores 0.39—below nanoGPT and vLLM, whose audiences are orders of magnitude smaller (Figure 7 caption; full comparison in Appendix D). It circulated functionally (“the site has visa slots”) with no creator name attached in indexable form, and the gap is symmetric across English and Chinese prompts, so it is not a language artifact. Recognition measures indexed name presence, not utility; the two correlate wherever creators attach names to work and dissociate for utility-driven artifacts that spread anonymously.

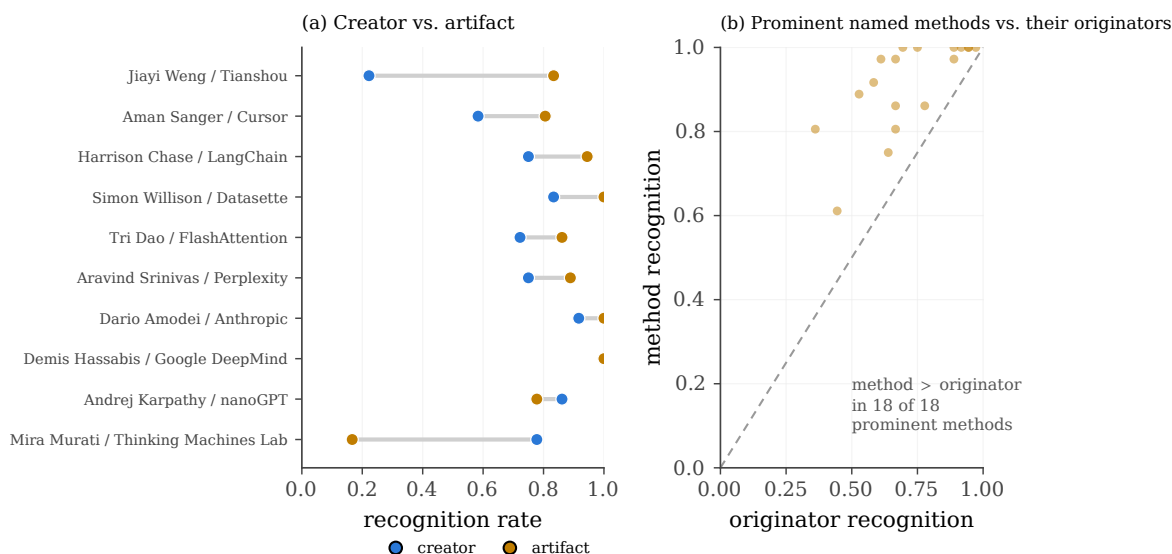


Figure 7. Named-artifact amplification. (a) Creator vs. artifact recognition for the 10 resolvable pairs (blue creator, amber artifact), sorted by artifact–creator delta: for independent creators the artifact uniformly out-ranks its maker; the exceptions are senior leaders. **(b)** Among named methods prominent enough to be standalone entities, the method out-ranks its originator in almost every case.

6 What Predicts Recognition

Turning from named artifacts to measurable correlates: what, among the quantities institutions actually track, predicts whether a name has propagated?

6.1 No bibliometric predicts recognition well

On the working-researcher cohort (603 researchers with OpenAlex bibliometrics), recognition rises with bibliometric productivity (Figure 8), but no single measure dominates: $\log_{10}(h\text{-index})$, raw citations, and attribution-weighted variants all explain around a fifth of the variance ($R^2 \approx 0.22$; Appendix E), and the differences among them are within noise. The headline is the ceiling, not the ranking: even the best bibliometric leaves roughly three-quarters of recognition variance unexplained. That residual is exactly what this instrument exists to measure—the part of recognition that flows not through the publication system but through the named-artifact and corpus-density channels of the surrounding sections. A bibliometric can rank researchers within the academy; it cannot stand in for whether a name has propagated.

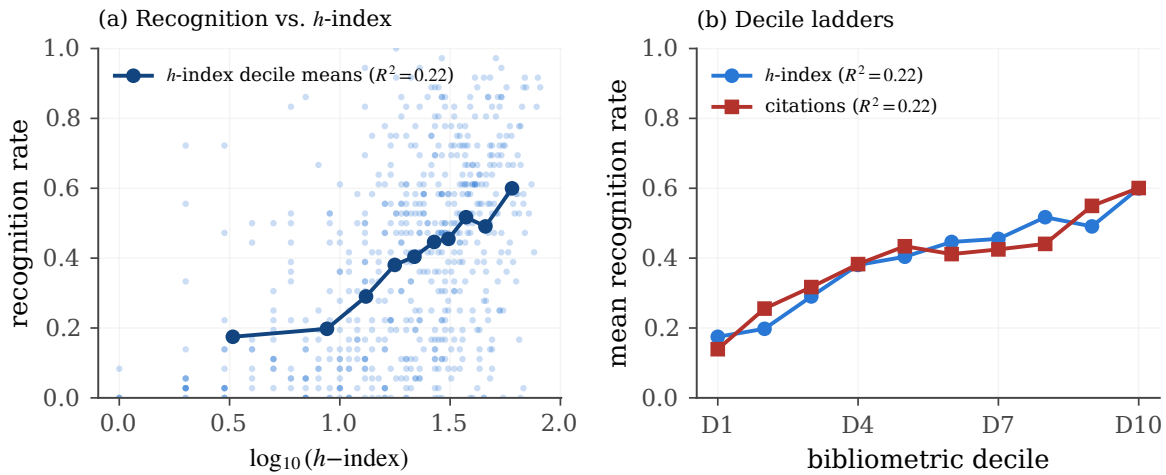


Figure 8. Bibliometric predictors. (a) Recognition vs. $\log_{10}(h\text{-index})$ for OpenAlex researchers, with decile means. (b) Decile ladders for $h\text{-index}$ and raw citations: both rise with productivity, neither dominates, and both leave most variance unexplained.

6.2 The corpus-density gradient: institutions and countries

Recognition varies with where a researcher works, and—crucially—not only through productivity. Among CS faculty, top-corpus-density institutions lead: an MIT faculty member scores 0.79, Berkeley 0.76, against lower-corpus-density peers (Figure 9a). The clean test holds citations fixed: restricting to a matched citation window (Figure 9b), MIT and Berkeley faculty still out-recognize UCSD and Irvine faculty at the *same* citation level—so the gradient is corpus density, not research output. Aggregated by country (Figure 10), the firm, well-sampled contrast is the USA (0.64, $n = 206$) above China (0.33) and India (0.26), whose confidence intervals do not overlap the USA’s; small-sample leaders are read as suggestive only. The mechanism is English-corpus density: identical research output propagates through a denser English-language ecosystem of social media, newsletters, podcasts, and tutorials for a US-based researcher than for a peer whose output travels mainly through the papers themselves.

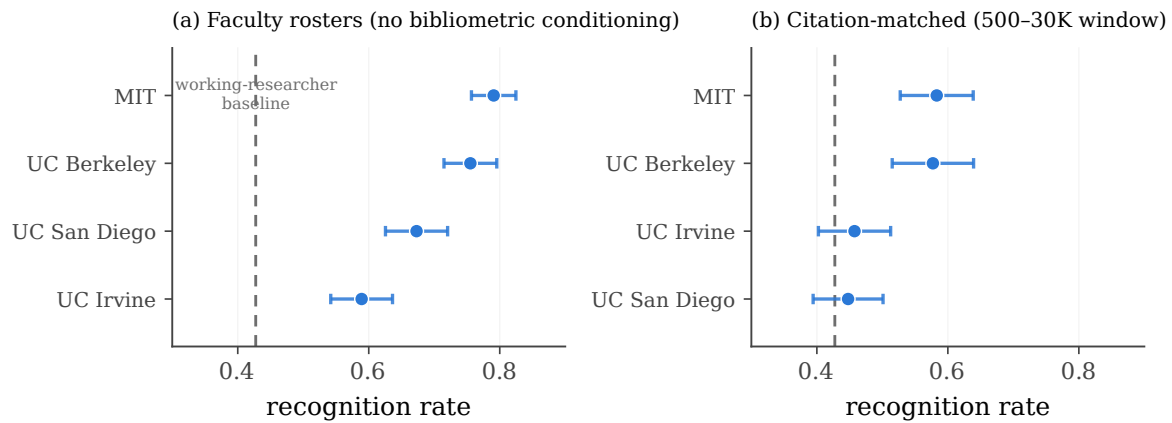


Figure 9. Institution gradient, two designs. (a) CS-faculty recognition by university (CSRankings rosters, no bibliometric conditioning). (b) At a matched citation window, top-corpus-density institutions still lead—corpus density, not productivity. Dashed line: working-researcher baseline.

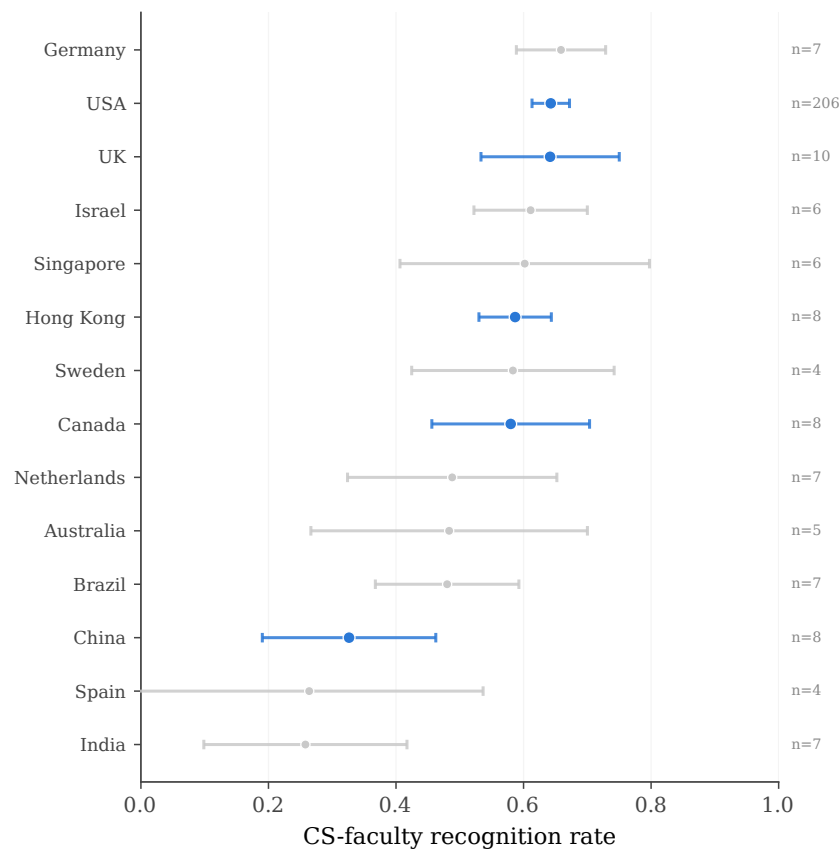


Figure 10. Country gradient in CS-faculty recognition (dot, 95% CI; faded rows $n < 8$, suggestive only). The USA leads China and India with non-overlapping intervals; the gradient tracks English-corpus density, not research quality.

6.3 The cleanest predictor: a recorded attention ledger

Every predictor so far proxies corpus exposure after the fact. News events invert the problem: the attention the world paid an event is itself a public, time-stamped record, so here the exposure side of the corpus-propagation account can be measured directly rather than inferred. We built a calibration cohort of 258 news events from 2021–2023, each carrying its English-Wikipedia daily pageview series, and probed them with the unchanged pipeline (Appendix H). Recognition rises log-linearly in total first-year attention. The decomposition is the sharp result: total attention factors into *peak* $\times$ *duration*, and *peak* dominates (standardized coefficient $+0.13$ against $+0.04$ on duration; duration adds a marginal R^2 of only 0.045, against 0.34 explained by peak alone). How *loud* a story got, not how *long* it lasted, drives recognition—because a story’s peak marks how many name-bearing documents were minted at climax, while a long tail of readership mints little additional text. Recognition follows document production, not audience retention; it is the temporal echo of the named-artifact mechanism, and it runs opposite to the *h*-index intuition that persistence should matter (Figure 11).

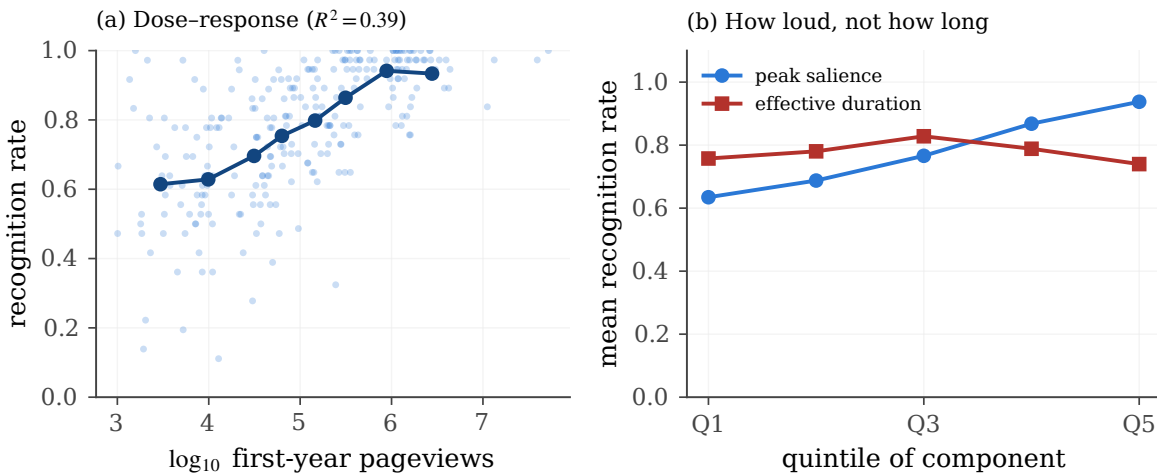


Figure 11. The attention ledger (258 news events, 2021–2023). **(a)** Recognition vs. $\log_{10}$ first-year pageviews, with decile means. **(b)** Quintile ladders for the two components of total attention: peak salience rises, effective duration is flat—recognition follows how loudly a story peaked, not how long it lasted.

7 Robustness and Validation

The findings above rest on three design choices—an LLM judge over embedding similarity, a binary recognition verdict over a graded score, and open-ended over closed-factoid probes—and on the headline numbers not being artifacts of wording, model vintage, judge family, or gold-answer construction. Table 1 summarizes the validation battery; Appendices J–O give the full evidence.

Three of these checks carry most of the weight and deserve the numbers the table compresses. First, *the scoring rule is load-bearing*: the entity factor explains roughly six times the variance the model factor does, and a disjoint replacement panel reproduces the event ordering, so the score measures the entity, not the fleet; the recognition verdict is what dis-

Table 1. The validation battery: each threat to the findings, the check run against it, and the outcome. Full tables and figures in the referenced appendices.

Threat	Check	Outcome	
Score reflects panel composition, not the entity	Variance decomposition over all verdicts; disjoint-panel re-score of the event cohort	Entity variance $\approx 6 \times$ model variance; 2026 replacement models reproduce the event ordering	App. J
Fluent hallucinators inflate scores	Recognition verdict on the full panel	The verdict scores a confident wrong-person biography as non-recognition; the zero-refusal fluent-hallucinator cluster contributes nothing	App. J
A judge-free embedding score would suffice	Judge-vs-embedding comparison	Embedding similarity is flat across recognition and credits hallucinations	App. J
A self-report score would suffice (just ask the model)	Pairwise self-report probe on three frontier models	Self-report reads the shared corpus prior, not the model’s own knowledge state; usable as a prior, not a measurement	App. I
Findings are an artifact of probe wording	Four-template paraphrase re-probe	Template explains $\approx 0\%$ of verdict variance; cohort ladder preserved ($\rho \geq 0.997$), only absolute levels shift	App. K
Findings are an artifact of probe language	Chinese-translated re-probe of a 240-entity subset	No cohort’s rate moves by more than 0.04; recognition is stored language-independently	App. G
The silent zone is corpus timing, not obscurity	Training-cutoff natural experiment	Matched difference-in-differences negative—recency cohorts gain <i>less</i> than capability drift; silence is intrinsic	App. L
Findings depend on the judge’s model family	Three-family re-judge	Cohort ranking identical under all three judges; GPT-5.1 stricter overall but a uniform offset moves no relative finding	App. M
Unknown names could score above zero	Synthetic-null entities of each gold recipe through the full pipeline	Floor ≈ 0.01 for people/artifacts, ≈ 0.06 for descriptive paper titles; floor-adjusting does not close the credential gap	App. N
Gold construction, context leakage, Wikipedia presence, or web attention drive the results	Dedicated ablations and regressions; long-window pageview baseline	All rejected; recognition is not a Wikipedia-page flag nor pageview rank	App. O

ciplines the zero-refusal fluent-hallucinator cluster, which an embedding score would credit as recognized (Appendix J). Second, *the silent zone is real obscurity, not corpus timing*: on two contemporaneously-published author cohorts, crossing a model’s training cutoff raises the recency cohorts *less* than it raises stable controls (matched difference-in-differences -0.24 and -0.17), even as a $\sim 0.1\text{--}0.3/\text{yr}$ capability drift accumulates across vintages—so absolute levels are comparable only within a run, but the silence itself is intrinsic (Appendix L), as the transformer fact-storage literature predicts [Geva et al., 2021, Petroni et al., 2019, Kandpal et al., 2023, Zhang et al., 2025]. Third, *floors and confounds hold*: floor-adjustment against the synthetic-null recipe leaves every credential below the working-researcher baseline, and dedicated ablations reject both Wikipedia presence and web-attention rank as alternative explanations (Appendices N, O). Orderings survive paraphrase and judge family (a three-family re-judge reproduces the cohort ranking under Gemini, Claude, and a uniformly stricter GPT-5.1); only absolute levels shift, so NameRank values are read as within-run relative gaps. Two further checks were run as standalone experiments and get their own treatment below: whether a model can simply be *asked* how well it recognizes an entity, and whether the score survives a change of probe language.

7.1 Self-report is not a shortcut for measurement

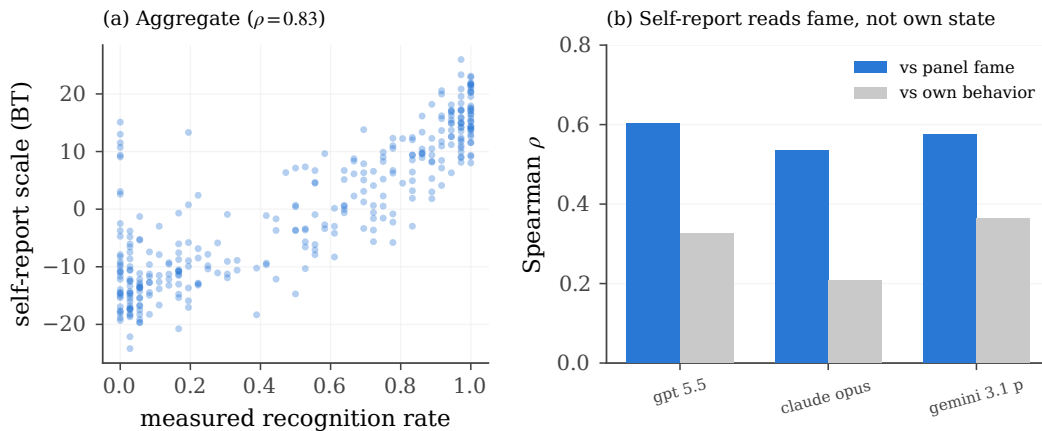


Figure 12. Introspection reads the corpus prior. (a) Bradley–Terry self-report scale vs. measured panel recognition: the ordering is recovered fairly well in aggregate. (b) But per model, self-report tracks cross-panel fame far better than the model’s own behavioral record—it reads what everyone knows, not its own weights.

Could a model simply be asked how well-known an entity is? We put this to a pairwise test: models rank which of two entities they recognize better, and we fit a Bradley–Terry scale from the comparisons (Appendix I). In aggregate the self-report scale recovers the panel recognition ordering reasonably well (Spearman $\rho \approx 0.83$; Figure 12a)—but the recovery is borrowed, not introspective. Two diagnostics show why. First, each model’s self-ranking tracks how *widely* a name is known across the panel ($\rho \approx 0.6$) far better than its own behavioral record on the entities it demonstrably knows ($\rho \approx 0.3$; Figure 12b). Second, the three vendors’ self-report scales agree with *each other* ($\rho \approx 0.90$) more than any agrees with the ground-truth panel recognition—they are reading a shared corpus prior, not their own weights [Kadavath et al.,

2022, Yin et al., 2023]. In one direction the claims are still trustworthy: opposite a name the model genuinely knows, the fictional side of a trap pair was almost never preferred. Self-report errs by reciting consensus fame, not by fabrication—usable as a prior, insufficient as measurement.

7.2 Recognition is prompt-language-invariant

Re-probing a 240-entity subset with a Chinese-translated template moves no cohort's recognition rate by more than 0.04 (Appendix G). This is the expected behavior of a recognition verdict: because the judge credits only a specific non-guessable fact, and whether the model stores that fact is a property of its weights rather than of the language it is queried in, the wording of the probe does not change what the model knows. An earlier graded-coverage version of this sub-run showed a sizeable Chinese-prompt lift for Chinese-named cohorts, but that was language-matched *verbosity*—a Chinese prompt draws a longer Chinese answer that accrues more coverage points—not extra knowledge; the recognition verdict removes it. A residual nudge (at most ~ 0.14) survives only for low-recognition Chinese-named credential holders, where a Chinese prompt occasionally tips a borderline model over the recognition threshold. NameRank is therefore comparable across prompt languages; all headline results use English prompts.

8 Discussion

Three questions organize what the findings imply beyond the measurements themselves: what *clock* the score reads, what it says to anyone who wants their own name to propagate, and what structural inequalities it exposes. We then close with limitations and a set of falsifiable predictions for future re-runs.

8.1 What Clock Does NameRank Read? Lifetime vs. Current-Term Recognition

A recognition score could measure two different quantities: *cumulative* presence—everything a name has deposited in the corpus over a career—or *current-term* salience, what the world is paying attention to now. Three findings established independently above jointly place NameRank in the first category. Within IMO gold medalists, medal year is essentially uncorrelated with the score (Section 4): recognition neither fades for older medalists nor spikes for recent ones. The marquee awards show the complementary shape: recognition does not decay with time since the prize—decades-old Nobel, Turing, and Fields laureates remain saturated rather than fading (Appendix B)—which rules out a *flow* reading, under which old laureates would have faded, even though the saturated marquee tier cannot itself trace the accrual slope (that is left to the unsaturated cohorts; Appendix B). And on news events, where attention is directly recorded, the score loads on *peak* salience rather than persistence (Section 6.3): what matters is how many name-bearing documents were minted, not how long the audience stayed. Read together: NameRank is a cumulative ledger of name-bearing text, snapshotted through the current model fleet—a lifetime measure weighted by documentation density, not a barometer of current attention. Current-term change is accessible only longitudinally, as within-run relative gaps across successive fleets (Section 8.5).

8.2 What Raises Recognition—and Whether Going Viral Helps

Read prescriptively, the findings converge on a single lever: recognition accrues to *distinctively-named, indexable production accumulated over a career*. Volume of name-bearing works leads citation weight (*h*-index over raw citations, Section 6); one clearly-named artifact outperforms many papers (Section 5); and the credentials that propagate on their own are exactly those that carry a named artifact—best-paper awards, named methods, marquee prizes backed by a career of production—while a bare olympiad medal does not (Section 4.3). Because the ledger is cumulative (Section 8.1), the effective strategy is long-horizon and structural rather than about any single achievement: keep depositing documents that carry a distinctive, low-collision name into English-indexed text. One caveat is attribution-direction: recognition accrues to an artifact’s name first and to its creator only through the corpus’s attribution chain (Section 5), so a tool buried under generic corporate branding builds no personal recognition.

This also bounds what public attention buys. Going viral helps only insofar as the spike is *captured as durable, name-bearing documents*: on news events the score loads on peak salience over persistence (Section 6.3), so a burst that mints many name-attached records converts while ephemeral engagement that leaves none does not. The conversion is category-dependent: within celebrity cohorts pageviews track recognition closely, but for technical creators attention *flow* and corpus *stock* largely decouple (Appendix O; a low-traffic but corpus-saturated library can outscore a high-traffic recent model). Attention is thus a means, valuable only when it deposits distinctively-named text into the corpus—never an end in itself.

8.3 Structural Consequences: Corpus-Density Inequality and Inherited Bias

The corpus-density gradient of Section 6.2—top-density institutions above their peers even at matched citations, recurring at country scale with the USA-over-China-over-India ordering as its backbone—is a structural inequality in name propagation *per unit of research output*. It is downstream of training-pipeline indexing decisions rather than of research quality (sampling caveats and the researcher-mobility confound are tabulated in Appendix Q).

NameRank also inherits the demographic biases of its judge and probed models and measures them faithfully rather than correcting them: beyond the $\sim 2\times$ East-Asian-surname attenuation documented by Li [2026], man-coded names outscore woman-coded names by a significant +0.085 among CS faculty (directional-only in the smaller researcher cohorts; Appendix P). We restrict the reading to the research cohorts—the larger cross-profession gaps (actors +0.21, athletes +0.27) reflect real fame differences in the sample, not a NameRank artifact—and report the gap as an inherited corpus-and-judge property, not a calibration target, that downstream uses must account for.

8.4 Limitations

Appendix Q tabulates every caveat with its direction and mitigation. The material ones: absolute levels are wording-conditional while rankings are robust; the run uses a single judge family, checked against others; the marquee-award tier measures the fame the prize both marks and made, and the sampled fellowship cohorts are upper bounds; only within-run *relative* gaps are comparable across model fleets; and paper and competition cohorts carry a small residual synthetic-null floor that is subtracted where relevant.

8.5 Registered Predictions for Future Re-Runs

NameRank is a snapshot, so a re-run each release cycle yields a longitudinal trajectory. Because absolute levels drift across model fleets, we state predictions as within-run *relative* gaps (so the drift cancels) and register four falsifiable ones for the next fleet: (i) the IMO-gold-vs-working-researcher gap (currently ≈ -0.28) widens further within 12 months, as named-artifact-attached careers accumulate text while credential-only cohorts do not; (ii) the median artifact-minus-creator delta on independent-creator pairs grows within 18 months; (iii) the top-vs-lower-corpus-density institution gap narrows but does not close within 24 months as more non-English training data enters frontier models; (iv) on a re-run of the news-event cohort (Section 6.3)—the cleanest of the four, since its exposure side is frozen in the pageview archive—the recognition gap between older and newer events at matched attention shrinks as post-event derivative text accumulates.

9 Related Work

9.1 Relation to the Predecessor Instrument

The direct predecessor of this work is the recognition analysis in the IKP benchmark [Li, 2026, §5.7], which established the findings this paper builds on: on 345 CS-researcher probes, $\log(\text{citations})$ explains only $\sim 35\%$ of recognition variance, with the remainder attributed in a controlled audit to *named-artifact amplification*, *name uniqueness* (common East Asian surnames attenuate recognition by $\sim 2\times$), and *subfield-ecosystem density*; and on 557 Wikidata-grounded probes, domain-specific recognition multipliers trace to the structure of web discourse around each fact type rather than to entity prominence. IKP nonetheless treated recognition as a benchmark byproduct. NameRank differs in each dimension that matters for an impact metric: a $[0, 1]$ recognition rate over a 36-model panel (vs. a six-landmark tier ladder calibrated for parameter estimation); thousands of entities across dozens of cohorts spanning industry, OSS, and mid-tier figures (vs. 345 researchers and 557 Wikidata probes); open-ended probes scored by a per-model recognition verdict (vs. closed-factoid binary verdicts); *h*-index rather than raw citations as the validated bibliometric correlate; and cross-model variance retained as signal rather than absorbed into a tier label.

9.2 Scientometric Measures of Researcher Impact

The classical instruments for researcher impact—citation count, *h*-index [Hirsch, 2005], journal impact factor—measure intellectual reach within the academic publication system, and are increasingly understood as failing under pressures relevant here: authorship-list inflation and uneven attribution [Larivière et al., 2019], the *hyperauthorship* problem of pricing contributions to thousand-author papers [Cronin, 2001] (a GPT-4 co-author’s citation count is mechanically large but says nothing about name-recognition), prestige-hierarchy biases spanning an order of magnitude across institutional tiers [Wapman et al., 2022], and within-researcher variance that motivates decompositions like the *Q* parameter [Sinatra et al., 2016].

None of these instruments cross the academic boundary cleanly. Citation count does not measure an industry executive or a startup founder, and *h*-index does not measure the creator of a widely-used open-source tool (Section 3). Bespoke domain metrics (Google PageRank for websites, GitHub stars for OSS, Spotify monthly listeners for musicians, IMDb scores for

film) do not interconvert. Collective-attention metrics do—search volume, Google Trends, and Wikipedia pageviews place any entity on one axis, and pageviews in particular are an established salience proxy [Mestyán et al., 2013, Ripberger, 2011]—but they count queries and article traffic rather than verified knowledge, and they are empirically near-orthogonal to NameRank in the zone that matters: undefined for the majority of entities that lack an article, and only weakly correlated with recognition within the technical and research cohorts that carry the findings (Appendix O). We use attention records as calibration input on the event cohort (Section 6.3), not as the measurand: NameRank reads the *outcome* of the training pipeline—what deployed models can correctly state—rather than the exposure that feeds it. The cross-domain measurement of eminence has a long prior tradition in *historiometry* [Simonton, 1999], which quantifies the reputational standing of eminent individuals from archival traces; NameRank can be read as a historiometric instrument whose archive is the frontier-model training corpus rather than biographical dictionaries and citation indices. For impact assessment that spans academia, industry, OSS, and the long tail of solo operators, a cross-domain instrument is needed; NameRank is the first we are aware of that reads this archive directly.

9.3 LLM-as-Judge for Quality and Impact

Thelwall [2024, 2025b,a] use ChatGPT to score paper *quality* against the UK REF rubric; Kousha and Thelwall [2025] extend to societal influence; Liu et al. [2026] forecast citation counts. These differ from NameRank in two respects: they ask the LLM to *judge* quality, where we ask what the model has *absorbed*; and they average within a single model across iterations, where we average across the full 36-model panel, treating cross-model variance as signal.

A separate line asks whether models *know what they know*: self-assessed confidence is well calibrated on aggregate question answering [Kadavath et al., 2022] but degrades at the boundary of a model’s own competence [Yin et al., 2023], and entity popularity modulates factual reliability [Mallen et al., 2023]. Section 7.1 adds an entity-level counterpart: pairwise self-reported recognition, aggregated with the Bradley–Terry machinery familiar from Chatbot Arena [Chiang et al., 2024], tracks the corpus salience shared across vendors rather than the reporting model’s own verified knowledge.

The known LLM bias most relevant to our setting is the *Matthew effect* [Merton, 1968] as it propagates through LLM training corpora: frontier models recall already-highly-cited work disproportionately, inflating the rich-get-richer dynamic familiar from the bibliometrics literature. We inherit this bias and treat it as a known property of the measurand—NameRank measures *what frontier models have absorbed*, which is itself shaped by the Matthew dynamic. The bias is part of what we measure, not a flaw to be corrected.

9.4 Credentials and Long-Run Outcomes

A longitudinal literature finds that early mathematical talent and olympiad medals predict—and causally raise—later *academic* productivity: the 50-year SMPY study [Lubinski et al., 2020], a medal-cutoff natural experiment [Agarwal and Gaule, 2020], and work documenting substantial within-elite heterogeneity [Güllich et al., 2025]. Our credential-treadmill finding does not contradict this literature; it changes the dependent variable from academic production to LLM-mediated name recognition and finds that the credential per se carries little of it (Section 4). The credential still signals ability and motivation; only its role as a name-propagation channel has shrunk—in the language of signaling theory [Spence, 1973], the LLM-corpus channel transmits

not the credential itself but the named, indexable production that may follow it.

10 Conclusion

NameRank operationalizes the recognition-variance residual that Li [2026] characterized but did not measure: a cross-domain, hallucination-resistant recognition rate over a 36-model panel, whose findings each subvert a default expectation—olympiad credentials no longer propagate names, named tools out-rank their makers, and a model asked to rank its own recognition returns the shared corpus prior rather than its own knowledge. The broader claim is that human curiosity about people and artifacts is now re-mediated by a few frontier models, and whether a name has propagated into their training corpora is itself a measurable quantity—usable as a cross-domain impact metric, a longitudinal tracker of how discourse reaches training pipelines, and a normative indicator for anyone who wants their name to survive the transition. The instrument measures the post-bibliometric impact channel; what that channel rewards is not the credential you were awarded but the named, indexable artifact you built and shipped.

Acknowledgements

This paper was produced using Pine Copilot’s voice-directed *whisper coding* workflow [Pine AI, 2026], in which the authors specify, discuss, and review the work by voice while a coding agent—Claude Code with Claude Opus 4.8 and Claude Fable 5—carries out the planning, coding, experiments, and paper writing.

References

- Ruchir Agarwal and Patrick Gaule. Invisible geniuses: Could the knowledge frontier advance faster? *American Economic Review: Insights*, 2(4):409–424, 2020.
- Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot arena: An open platform for evaluating LLMs by human preference. In *ICML*, 2024.
- Blaise Cronin. Hyperauthorship: A postmodern perversion or evidence of a structural shift in scholarly communication practices? *Journal of the American Society for Information Science and Technology*, 52(7):558–569, 2001.
- Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. Knowledge neurons in pretrained transformers. In *ACL*, 2022.
- Roger R. Davidson. On extending the bradley–terry model to accommodate ties in paired comparison experiments. *Journal of the American Statistical Association*, 65(329):317–328, 1970.
- Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In *EMNLP*, 2021.

- Arne Güllich et al. Identifying high-performing individuals in talent domains: bounded learning curves, ceiling effects, and the limits of credentialing. *Science*, 2025.
- J. E. Hirsch. An index to quantify an individual’s scientific research output. *Proceedings of the National Academy of Sciences*, 102(46):16569–16572, 2005.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. Large language models struggle to learn long-tail knowledge. In *ICML*, 2023.
- Kayvan Kousha and Mike Thelwall. Using ChatGPT for societal-impact assessment of research. *Journal of the Association for Information Science and Technology*, 2025.
- Vincent Larivière, Cassidy R. Sugimoto, and Benoit Macaluso. Authorship inflation in scientific publishing: A bibliometric review. *Journal of Informetrics*, 2019.
- Kalev Leetaru and Philip A. Schrodt. GDELT: Global data on events, location, and tone, 1979–2012. In *ISA Annual Convention*, 2013.
- Bojie Li. Incompressible knowledge probes: Estimating black-box LLM parameter counts via factual capacity. *arXiv preprint*, 2026.
- Yuan Liu et al. Large language models predict highly cited papers from abstracts. *arXiv preprint 2601.13627*, 2026.
- David Lubinski, Camilla P. Benbow, and Harrison J. Kell. Sex differences in mathematical reasoning ability and longitudinal predictors of STEM careers: A 50-year follow-up of the study of mathematically precocious youth. *Psychological Science*, 2020.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *ACL*, 2023.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *NeurIPS*, 2022.
- Robert K. Merton. The Matthew effect in science. *Science*, 159(3810):56–63, 1968.
- Márton Mestyán, Taha Yasseri, and János Kertész. Early prediction of movie box office success based on wikipedia activity big data. *PLoS ONE*, 8(8):e71226, 2013.
- Fabio Petroni, Tim Rocktäschel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, Alexander H. Miller, and Sebastian Riedel. Language models as knowledge bases? In *EMNLP*, 2019.
- Pine AI. Pine AI: The most natural human-computer interface is your voice. Blog post, 2026. URL <https://www.19pine.ai/blog/pine-ai-the-most-natural-human-computer-interface-is-your-voice>. Accessed 2026-06-28.

- Joseph T. Ripberger. Capturing curiosity: Using internet search trends to measure public attentiveness. *Policy Studies Journal*, 39(2):239–259, 2011.
- Adam Roberts, Colin Raffel, and Noam Shazeer. How much knowledge can you pack into the parameters of a language model? In *EMNLP*, 2020.
- Dean Keith Simonton. Significant samples: The psychological study of eminent individuals. *Psychological Methods*, 4(4):425–451, 1999.
- Roberta Sinatra, Dashun Wang, Pierre Deville, Chaoming Song, and Albert-László Barabási. Quantifying the evolution of individual scientific impact. *Science*, 354(6312):aaf5239, 2016.
- Michael Spence. Job market signaling. *The Quarterly Journal of Economics*, 87(3):355–374, 1973.
- Mike Thelwall. Can ChatGPT evaluate research? a test of REF 2021 article quality scores. *arXiv preprint 2402.05519*, 2024.
- Mike Thelwall. Multi-iteration averaging for LLM-as-judge research evaluation. *arXiv preprint 2506.13525*, 2025a.
- Mike Thelwall. ChatGPT research-quality scoring across academic disciplines. *arXiv preprint 2504.04464*, 2025b.
- K. Hunter Wapman, Sam Zhang, Aaron Clauset, and Daniel B. Larremore. Quantifying hierarchy and dynamics in US faculty hiring and retention. *Nature*, 610:120–127, 2022.
- Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. BGE: One-stop retrieval toolkit for search and RAG, 2024.
- Zhangyue Yin, Qiushi Sun, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Xuanjing Huang. Do large language models know what they don’t know? In *Findings of ACL*, 2023.
- Yongqi Zhang et al. The law of knowledge overshadowing in large language models. *ACL*, 2025.

The appendices are organized in three blocks. Appendix A completes the instrument specification of Section 2. Appendices B–I give the full tables and case detail behind the findings of Sections 3–7, roughly in the order they appear. Appendices J–O give the evidence behind the validation battery of Section 7, one appendix per threat; Appendices P and Q cover the inherited-bias discussion and the limitations summary.

A Probe, Panel, and Cohort Specification

Why disambiguation does not contaminate scoring. The probe context names role, affiliation, and field—never specific contributions, papers, dates, or named artifacts—and the recognition judge credits only facts that go *beyond* what the context supplies (Section 2.4). A model that recognizes the entity states specific facts the context did not give; a model that does not produces generic statements about the role, which earn no recognition. The sharpest empirical check comes from the GPT-5 system-card author cohort, whose disambiguating context is identical across all 80 entities and which scores at the synthetic-null floor: the panel withholds recognition rather than manufacturing a generic in-context biography. A dedicated context-ablation (Appendix O) confirms the same conclusion.

Table 2. The 36-model panel, by vendor class. Thinking-mode variants are used wherever available; the panel spans frontier-reasoning to small (sub-30B) classes.

Western (n=20)	Chinese (n=16)
GPT-5.3, GPT-5.4, GPT-5.5-think, GPT-OSS-20b-think	DeepSeek-V3.2-think, V4-flash-think, V4-pro-think
Claude Opus 4.6-think, Sonnet 4.6-think, Fable 5-think	Qwen3-8b-think, 32b-think, 235b-a22b-think, 3.5-397b-a17b-think, 3.7-Max-think
Gemini 2.5-pro-think, 3-flash-think, 3.1-pro	GLM-4.7-think, 5.1-think, 5.2-think
Llama-3.1-8b, 3.3-70b, 4-Maverick	Kimi-K2.6-think, K2.7-code-think
Mistral-Small-24b, Phi-4	MiniMax-M2.7-think, M3-think
Gemma-3-4b, 3-12b, 4-31b	Step-3.7-Flash-think
Nemotron-3-Ultra-think, Nano-30b-think	

Table 3. Core-cohort entities by top-level group (main dataset; 4,685 real entities plus 45 synthetic-null controls). The award-ladder, news-event, LLM-area, and university-baseline extensions are specified in Appendices B, H, and their own sections. Full per-cohort breakdown in the [released cohort specification](#).

Group	<i>n</i>	Representative cohorts
Credentialed individuals	801	IMO/IOI/CMO/NOI/CPhO medalists, Putnam, ICPC, Rhodes, MSRA Fellowship
Working academics	1,210	CS faculty; OpenAlex/IKP researchers with 500–30,000 citations
Industry figures	122	founders, VCs, YC companies, AI-policy officials
Named artifacts	1,115	foundation models, OSS projects, papers, benchmarks, named methods, datasets
Mid-tier cross-profession	1,214	writers, athletes, actors, chefs, journalists, podcasts, books
Diagnostic reference set	37	hand-curated public figures, engineers, and artifacts
Special cohorts	186	GPT-5 system-card authors, DeepSeek-V3 authors, conferences, awards

B Mid/Late-Career Awards as Recognition Predictors

This appendix gives the construction and full tables behind Section 4.3 and Figure 6.

Cohorts. Laureate rosters were pulled from Wikidata (award-received statements, with the award-date qualifier supplying the year): Turing Award (77 human laureates with an English label), Fields Medal (64), Nobel Prize in Physics restricted to 2000–2023 (64), Gödel Prize (80), ACM Prize in Computing (18), and, sampled to 60 each from their 2000–2023 Wikidata-listed recipients, the MacArthur Fellowship and ACM Fellow honors; the Sloan Research Fellowship (49 listed for 2000–2023) is the early-career anchor. The five prize cohorts (Turing, Fields, Nobel, Gödel, ACM Prize) are complete rosters by construction—every laureate has a Wikidata item. The three fellowship/honor cohorts (MacArthur, ACM Fellow, Sloan) are sampled from Wikidata’s coverage, which omits lower-profile recipients, so their means are *upper bounds* and are read as such.

Panel, golds, and contexts. These cohorts are probed on the same 36-model panel and scored by the same recognition judge as every other cohort. Contexts follow the credential convention—award name and year only, never contributions (e.g., “a mathematician who received the Fields Medal in 2014”). Gold answers are the laureate’s en-Wikipedia lead section where available, otherwise a factual award-and-affiliation record, so recognition is credited only for facts the model states beyond the award itself.

Table 4. The award ladder under the recognition metric. Recognition is the panel fraction; *vs. base* is relative to the working-researcher baseline (recognition 0.46 on this run’s matched $n=60$ subsample and the IMO control 0.19 on $n=40$ —both slightly above the headline full-cohort values of 0.40 and 0.12 used elsewhere, and on the same panel and judge). The three marquee late-career prizes and the ACM Prize saturate near the ceiling; the sampled mid-career honors sit high but are upper bounds; the complete-roster mid-career prizes (Gödel) and the LLM-area author cohorts sit between the honors and the baseline; the early olympiad credential (IMO) sits well below it.

Award	Stage	n	Recognition	vs. base
Nobel Prize in Physics (2000–23)	late	64	0.98	+0.52
Turing Award	late	77	0.97	+0.51
Fields Medal	late	64	0.96	+0.50
ACM Prize in Computing	mid	18	0.95	+0.49
MacArthur Fellowship [‡]	mid	60	0.85	+0.39
Gödel Prize	mid	80	0.81	+0.35
ACM Fellow [‡]	mid	60	0.79	+0.33
Sloan Research Fellowship [‡]	early	49	0.78	+0.32
Named-method originators	mid	84	0.65	+0.19
Best-paper award authors	mid	65	0.53	+0.07
Foundational-paper authors	mid	85	0.45	+0.00
OpenAlex researchers (baseline)	—	60	0.46	—
IMO gold (control)	early cred.	40	0.19	−0.27

[‡] Wikidata-coverage sample (omits lower-profile recipients); cohort mean is an upper bound.

The career arc, in one table. Every cohort with a career of named production behind it clears the working-researcher baseline, and the ordering traces the paper’s thesis. Two reading notes keep it honest. First, the individual laureate cohorts saturate near the ceiling—a Nobel, Turing, or Fields laureate is an individually famous person the judge recognizes directly—so

the marquee tier measures fame the prize both marks and made, not the prize in isolation. Second, three fellowship cohorts (MacArthur, ACM Fellow, Sloan; marked ‡) are sampled from Wikidata, which lists only recipients prominent enough to have an article, so their means are upper bounds and should be read as such. The cleanest comparison is therefore among the *complete-roster* cohorts evaluated on identical rosters: the early olympiad credentials (below baseline), the foundational- and best-paper author cohorts and named-method originators (at-to-above baseline, rising with how directly a named artifact attaches to the person), and the complete-roster prizes (Gödel, ACM Prize, and the marquee three) at the top. Along that clean spine, recognition rises with named-artifact density, exactly as the mechanism predicts.

Saturation, not a lifetime clock. The marquee prizes cannot themselves reveal the accrual clock, because they sit at the ceiling: a Nobel, Turing, or Fields laureate is recognized by essentially the whole panel (0.96–0.98), so there is no headroom for recognition to “keep rising” with time since the award. Pooling the three marquee cohorts and standardizing recognition against each award’s own mean, the trend against years-since-award is flat (slope ≈ 0.00 standardized units/yr; non-monotonic across decade bins), as expected at saturation. The cumulative-ledger reading of Section 8.1 is therefore carried not by the saturated marquee tier but by the *unsaturated* cohorts, where recognition has room to track accumulated named production: the cohort ladder itself (below), the training-cutoff gradient (Appendix L), and the NOI case (Appendix C), where post-medal careers—not the medal—separate the recognized from the silent.

C Medal Tiers: The Complete NOI 2009 Roster

The credential ladder above measures *gold* medalists only, so it cannot say whether the medal’s grade carries signal. To test this we probed the complete medal roster of a single contest — NOI 2009: every gold, silver, and bronze medalist (103 in all) — on the same 36-model panel and recognition judge as every other cohort. The roster comes from the official CCF medal list, cross-validated line-by-line against the community OIerDb database (identical names, schools, and scores); the released files include each medalist’s contest score and rank. Probe names carry the Chinese characters alongside the romanization, which — together with the year-and-school context — disambiguate medalists whose Pinyin collides with common names.

Measurement protocol. This roster is scored under the recognition judge used throughout the paper (Appendix J). Each medalist’s gold answer is an individual, web-grounded profile of who the person actually is — their real career and named contributions, retrieved by a search-augmented model and length-normalized to ~ 55 words. The judge is shown the probe context and returns a binary *recognized* verdict: it credits recognition only when the response states a specific, non-guessable fact about the person that is true either per the gold or per the judge’s own knowledge, and *not* derivable from the disambiguating context, so a response that merely restates “a 2009 NOI gold medalist from [school]” is not recognition. Because the entities are competition participants—for whom a judge has no reliable per-individual memory—the anti-confabulation provision applies: fine-grained minutiae asserted via the judge’s own knowledge (other-year participations, exact scores) are not credited, only gold-present facts or major documented achievements. The per-entity score is the fraction of the 36-model panel that recognizes the person.

Most of the population sits near the floor, and that is the finding. The typical NOI

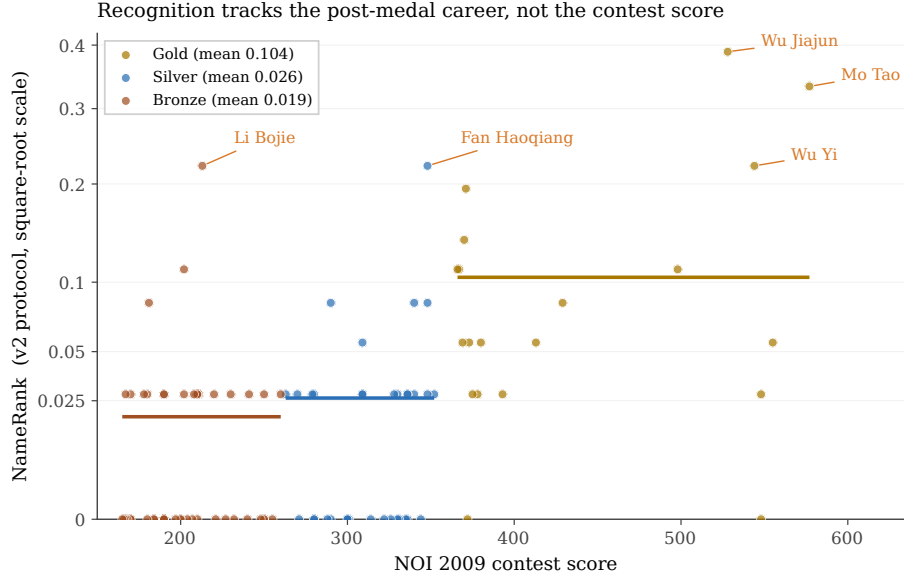


Figure 13. NameRank against NOI 2009 contest score for the complete medal roster (square-root y -axis, so the dense near-zero band and the recognized tail share one panel), with within-tier means as horizontal lines. Most medalists are silent (≈ 0); only the five recognized medalists above 0.2 are labelled. Several golds scoring ≥ 500 stay near the floor unlabelled, while two much lower-scoring medalists (Fan Haoqiang, Li Bojie) are recognized—recognition tracks the post-medal career, not the contest score.

Table 5. NOI 2009 medal tiers under the recognition judge. The score is the panel recognition rate—the fraction of the 36 models that demonstrate genuine recognition (Appendix J); “silent” counts medalists no model recognizes, “recognized” those at least one model does. The protocol-matched synthetic-null floor is 0.009 (three fictional NOI medalists, same recipe), so the gold-tier mean is almost entirely above-floor signal. Kruskal–Wallis across tiers $H = 25.9$ ($p < 10^{-5}$); gold vs. silver Cliff’s $\delta = +0.62$; silver vs. bronze indistinguishable ($p = 0.27$, $\delta = +0.13$).

Tier	n	Mean	Median	Silent (0 models)	Recognized (≥ 1)
Gold	20	0.104	0.056	2/20	18/20
Silver	34	0.026	0.028	16/34	18/34
Bronze	49	0.019	0.000	28/49	21/49

medalist is recognized by only a handful of the 36 models: the gold-tier median is 0.056 (about two models), the silver median 0.028, the bronze median 0 (Table 5, Figure 13). Even in the gold tier, where 18 of 20 are recognized by at least one model, the mass is a thin near-floor band with a short recognized tail—a frontier panel stores only a small, documented slice of even this pre-selected population. This is the silent-not-misleading property (Section 3) operating at full strength: against a synthetic-null floor of 0.009, the gold mean of 0.104 is a real above-floor signal, and the silver/bronze means (0.026, 0.019) sit just above it.

The gold/non-gold cliff survives; silver and bronze do not separate. The gold tier stands clear of the lower tiers (Cliff’s $\delta = +0.62$ vs. silver, Kruskal–Wallis $p < 10^{-5}$), while silver and bronze are statistically indistinguishable ($\delta = +0.13$, $p = 0.27$). The ~ 20 -person national gold roster is the documented one—admissions coverage, competitive-programming lore, team-selection reporting—and it carries by far the most recognition mass (mean 0.104 against

0.026 and 0.019). Contest score *within* a tier does not predict recognition (Figure 13): the effect is the roster cut, not the marginal point.

The recognized names are career-driven — the treadmill again. The medalists who rise above the near-floor band are exactly those whose *post-medal* careers produced named, indexed work: the most-recognized medalist overall is a gold who is now a Stanford computer-vision professor (Wu Jiajun, 0.39); the most-recognized silver matches the gold-tier top on a prominent vision-research career (Fan Haoqiang, 0.22, later an IOI gold and a Megvii research lead); and the most-recognized bronze ranks near the bottom of the contest’s score column yet ties the top silver (Li Bojie, 0.22, the author of this paper). The medal grade sets the odds of entering a documented cohort; the recognition itself tracks what the person later built.

The noise floor is calibrated with protocol-matched synthetic-null entities—three fictional NOI 2009 medalists built on the same gold recipe and scored by the same judge—and comes out near zero (0.009): the anti-confabulation provision leaves fictional competitors essentially no recognized facts. The gold-tier mean of 0.104 is therefore an above-floor signal by an order of magnitude, and the silver (0.026) and bronze (0.019) means are small but sit clearly above the floor.

D Artifact–Creator Pairs: Inversion, Injection, and Asymmetry Detail

This appendix gives the full tables behind Figure 7 and the Tianshou case study (Section 5).

D.1 The inversion and injection tables

Table 6. The artifact-over-creator inversion under the recognition metric. NR_c and NR_a are creator and artifact recognition; $C \rightarrow A$ is the fraction of responses about the creator that mention the artifact, $A \rightarrow C$ the reverse. Boldface: artifact exceeds creator. For independent creators of a single artifact (top block) the artifact uniformly out-ranks its maker; the exceptions (bottom) are senior leaders with independent recognition.

Creator	NR_c	Artifact	NR_a	$\Delta(a-c)$	$C \rightarrow A$	$A \rightarrow C$
Jiayi Weng	0.22	Tianshou	0.78	+0.56	0.25	0.00
Aman Sanger	0.58	Cursor	0.81	+0.23	1.00	0.00
Tri Dao	0.72	FlashAttention	0.94	+0.22	0.74	0.54
Harrison Chase	0.75	LangChain	0.94	+0.19	1.00	0.22
Simon Willison	0.83	Datasette	1.00	+0.17	0.54	0.89
Aravind Srinivas	0.75	Perplexity	0.89	+0.14	1.00	0.33
Dario Amodei	0.92	Anthropic	1.00	+0.08	1.00	0.41
Demis Hassabis	1.00	Google DeepMind	1.00	0.00	1.00	0.62
Andrej Karpathy	0.86	nanoGPT	0.78	−0.08	0.05	0.76
Mira Murati	0.78	Thinking Machines Lab	0.17	−0.61	0.97	1.00

The inversion is observational, not inducible. Under a graded coverage score, an earlier version of this experiment showed an apparent positive lift—but that lift was the model echoing the injected artifact name back into a response that then earned coverage credit, not new stored knowledge. The recognition judge closes exactly that channel: the injected clause is part of the context, so a response that merely restates it earns nothing, and only facts the model knew independently count. Scored this way, the mean lift is essentially zero (−0.01), positive

Table 7. Context-injection experiment. Δ is the creator’s recognition change when the artifact name is added to the disambiguating context (configuration B – configuration A). Under the recognition judge, which credits only facts beyond the injected clause, the lift is essentially zero: naming the artifact cannot manufacture recognition of the creator. Weak positive lift appears only for the two least-known creators.

Creator	Artifact injected	Δ
Jiayi Weng	Tianshou	+0.11
Simon Willison	Datsette	+0.08
Tri Dao	FlashAttention	+0.06
Aman Sanger	Cursor	+0.03
Harrison Chase	LangChain	0.00
Demis Hassabis	Google DeepMind	0.00
Andrej Karpathy	nanoGPT	−0.03
Mira Murati	Thinking Machines Lab	−0.08
Aravind Srinivas	Perplexity	−0.08
Dario Amodei	Anthropic	−0.11
Lilian Weng	lilianweng.github.io	−0.11
Mean over 11 pairs		−0.01

only for the two least-known creators (Jiayi Weng, Tri Dao) and negative or null everywhere else. Naming the artifact does not manufacture recognition of the creator. The artifact-over-creator inversion is therefore a property of how attribution is distributed in the corpus—a probe about a famous artifact often omits its creator—not an effect a prompt intervention can induce, which is the stronger and cleaner reading of the mechanism.

D.2 The Tianshou case

Table 8. The Tianshou case: Jiayi Weng against OpenAI individual-contributor peers who share the affiliation but not a widely-named artifact. The point is the artifact-mediation gap: his tool Tianshou vastly out-ranks him, and the peers who lack such an artifact cluster near the floor. (Trevor Blackwell, a YC co-founder, carries independent recognition and is not an anonymous-IC comparison.)

Entity	Recognition
Jiayi Weng	0.22
Tianshou (his artifact)	0.78
Vicki Cheung	0.39
Pamela Vagata	0.33
Alex Wei	0.11
Andrea Vallone	0.08
Adam Fry	0.03
Aaditya Singh	0.00
Peer mean (anonymous ICs)	0.16

The case distills the mechanism to one person: Jiayi Weng’s recognition (0.22) sits in the range of his OpenAI IC peers, but his named artifact Tianshou (0.78) vastly exceeds him—the recognition attaches to the tool, and reaches the person only through it. The peers who never

shipped a widely-named artifact cluster near the floor. Per-model mention data for all 11 verified pairs (fraction of non-refusal responses that name the counterpart) is released as a supplementary CSV; the models that recognize Jiayi Weng are exactly those whose response names Tianshou.

The boundary: reach without name propagation (Section 5.3). Probed on the same panel, a uniquely-named, GitHub-documented CTF (Hackergame) out-recognizes a course-review site (icourse.club) and the visa-monitoring site tuixue.online, all far below the named-artifact anchor nanoGPT. The ordering tracks whether the site’s name co-travels with documented, English-readable text, not its user base—the same attribution mechanism, at the boundary where an artifact spreads functionally but anonymously.

E Bibliometric Predictors

Full bibliometric-regression results for NameRank on the OpenAlex long-tail researcher cohort ($n = 603$ with complete bibliometrics, citations 500–30,000). `fractional_citations` divides each paper’s citations by its author count; `first_last_citations` counts only papers where the researcher is first or last author; `n_works` is the raw count of works; `mean_authors` is the mean author count per paper. All R^2 on $\log(1 + x)$ -transformed predictors.

Table 9. Sole-predictor R^2 on recognition. No single bibliometric dominates: h -index, raw citations, and the attribution-weighted variants (fractional and first/last-author citations) all cluster around a fifth of the variance, and the differences among them are within noise. Author count carries no signal, and even the best predictor leaves most of the variance unexplained.

Predictor	Sole R^2
$\log(\text{fractional citations})$	0.26
$\log(\text{first/last-author citations})$	0.24
$\log(h\text{-index})$	0.22
$\log(\text{raw citations})$	0.22
$\log(n_{\text{works}})$	0.16
$\log(\text{mean authors/paper})$	0.00

The predictors are close substitutes: attribution-weighted fractional citations edge ahead, but h -index and raw citations tie near $R^2 \approx 0.22$, and the joint citation-plus- h -index model reaches only 0.28—each adds ≈ 0.06 beyond the other because the two are correlated ($r \approx 0.6$). Recognition rises near-monotonically across h -index deciles (Table 10), from 0.18 at the bottom decile to 0.60 at the top, but never approaches a ceiling that would let a bibliometric stand in for recognition: every predictor leaves roughly three-quarters of the variance unexplained. That residual is what the named-artifact and corpus-density channels (Sections 5, 6.2) carry, and it is why bibliometrics alone cannot serve as a recognition proxy.

F Institution and Country Gradient Detail

Table 11 gives the institution-level view; Table 12 the country-level aggregation behind Figure 10. The matched-citation companion design (Figure 9) separates corpus density from productivity.

Table 10. Mean recognition rate by h -index decile within OpenAlex long-tail researchers. The ladder rises near-monotonically from the bottom decile to the top; the bottom decile is floored at the working-researcher floor, not at zero.

Decile	Mean h	Mean recognition
D1	4	0.18
D4	18	0.38
D7	31	0.46
D10	61	0.60

Table 11. CS-faculty recognition by institution (institutions with $n \geq 8$ matched faculty). The top-corpus-density institutions lead; the gradient recurs at country level (Figure 10) and holds at matched citations (Figure 9).

Institution	n	Recognition
Stanford University	13	0.78
University of Washington	21	0.74
Carnegie Mellon University	50	0.71
Princeton University	15	0.67
Cornell University	17	0.65
University of Michigan	20	0.62

The matched-citation design. The faculty-roster comparison above cannot separate corpus density from productivity, because it does not condition on output. A companion design does: sampling researchers into a fixed citation window (500–30,000) stratified by institution, so the compared groups have matched bibliometrics by construction. At matched citations the top-corpus-density institutions still lead—MIT and Berkeley faculty out-recognize UCSD and Irvine faculty at the same citation level (Figure 9b)—so the gradient is corpus density, not research output.

Table 12. CS-faculty recognition by country of institutional affiliation, for countries with ≥ 8 sampled faculty. The firm, well-sampled contrast is the USA against China (non-overlapping intervals); the mid-sized samples (UK, Hong Kong, Canada) sit between them and should not be finely ranked.

Country	n	Recognition
USA	206	0.64
UK	10	0.64
Hong Kong	8	0.59
Canada	8	0.58
China	8	0.33

The well-sampled backbone of the gradient is the USA (recognition 0.64) against China (0.33), whose 95% confidence intervals do not overlap. The mechanism is English-corpus density: identical research output propagates through a denser English-language ecosystem for a US-based researcher than for a peer whose output travels mainly through the papers themselves.

G Cross-Language Robustness

Sub-run cohort selection. The Chinese-prompt sub-run covers 240 entities: the full diagnostic reference set (37), all NOI China gold medalists (29), all DeepSeek-V3 paper authors (69), 30 random samples each from the CMO, CPhO, and MSRA Fellowship cohorts, and 5 each from the writer, filmmaker, and politician cohorts as Western controls— 240×36 records.

Table 13. Cross-language NameRank by cohort: $\Delta = \text{NR}_{\text{zh}} - \text{NR}_{\text{en}}$ on the 240-entity Chinese-prompt sub-run. Under the recognition verdict the prompt language barely moves the score—every cohort delta is within ± 0.04 —because recognition turns on whether the model can state a specific non-guessable fact, and that knowledge is language-invariant.

Cohort	n	NR_{en}	NR_{zh}	Δ
MSRA PhD Fellowship	30	0.166	0.183	+0.018
DeepSeek-V3 authors	69	0.069	0.075	+0.006
CMO China gold	30	0.029	0.028	−0.001
NOI China gold	29	0.086	0.084	−0.002
CPhO China first prize	30	0.034	0.029	−0.006
Diagnostic reference set	37	0.627	0.615	−0.012
Writers (control)	5	0.906	0.872	−0.033
Filmmakers (control)	5	0.994	0.961	−0.033
Politicians (control)	5	0.900	0.861	−0.039

The prompt-language effect is essentially null. This is itself a finding, and a reassuring one for a cross-lingual instrument: switching the probe from English to Chinese changes no cohort’s recognition rate by more than 0.04, and the Chinese-named credential cohorts (DeepSeek authors +0.006, NOI −0.002, CMO −0.001) that a language-matching artifact would most inflate move least. An earlier graded-coverage version of this sub-run showed a sizeable Chinese-prompt lift for Chinese-named cohorts; that lift was language-matched *verbosity*—a Chinese prompt elicits a longer Chinese answer that scores more coverage points—not extra knowledge. Once the judge credits only a verified non-guessable fact, the wording of the question stops mattering, because whether the model knows Lin Yankai is a property of its weights, not of the language it is asked in.

Per-model prompt-language deltas. The residual shift is small and not vendor-specific: the mean absolute per-model delta is 0.023, and the two model classes are indistinguishable in aggregate (Western −0.005, Chinese +0.002). The largest per-model movers—deepseek-v4-pro-think +0.08, gemma-3-12b +0.08; llama-3.1-8b −0.05, gemma-3-4b −0.04—are small-model noise on either side, not a systematic home-language advantage.

Per-entity deltas. The full per-entity $\Delta(\text{NR}_{\text{zh}} - \text{NR}_{\text{en}})$ table is released as a supplementary CSV. Even the extremes are modest—the largest suppressions are Harrison Chase −0.19, Aleksander Madry −0.14, and Canyi Lu −0.14, all English-context entities where a few borderline models drop the fact on the Chinese prompt:

Entity	NR _{en}	NR _{zh}	Δ	Cohort
Jiayi Weng	0.22	0.22	+0.00	reference set
Tri Dao	0.72	0.72	+0.00	reference set
Andrej Karpathy	0.86	0.89	+0.03	reference set
FlashAttention	0.86	0.92	+0.06	reference set
Tianshou	0.83	0.89	+0.06	reference set
Bojie Li	0.22	0.19	−0.03	reference set
Pan Zhou	0.08	0.22	+0.14	MSRA Fellowship
Lin Yankai	0.06	0.17	+0.11	NOI China gold
Chen Jianfei	0.03	0.14	+0.11	NOI China gold
Yingwei Pan	0.17	0.28	+0.11	MSRA Fellowship

The residual pattern, where it exists at all, is confined to *low-recognition* Chinese-named credential holders (bottom block): for a name the panel barely knows, a Chinese prompt occasionally tips a borderline model over the recognition threshold, worth at most ~ 0.14 . For every entity the panel actually recognizes—the reference set—the delta is within ± 0.06 . Name attribution that has genuinely propagated is stored language-independently; the prompt language cannot manufacture it and barely nudges it.

H News-Event Cohort: Construction and Detail

This appendix documents the construction of the 258-event cohort of Section 6.3 and reports the full tables behind its findings. All construction scripts, inputs, and derived data are released under `experiments/t4_1_news_events/` in the repository.

H.1 Sampling frame and filters

Candidate events are harvested from the “Events” sections of the English Wikipedia year articles (2021, 2022, 2023) and of 32 country-year articles (“2021 in India”, “2022 in Nigeria”, ...) chosen to balance nine world regions (2,245 unique linked articles). Each candidate is classified by Gemini 3 Flash Preview (the study’s judge family; temperature 0, structured output) from its title, short description, and first paragraph into: discrete event or not, start year and month, primary country, region, and category. An article qualifies as a *discrete event* if it describes a specific occurrence or tightly bounded series (an election, a storm, a trial, a tournament, an accident, a protest wave) that began at an identifiable date; people, organizations, products, laws, ongoing multi-year topics, and list/timeline articles are excluded (1,041 qualify). Post-classification filters require a standard-type article with an intro of ≥ 40 words and a first-year pageview record of $\geq 1,000$ total views over ≥ 200 recorded days; 655 events survive. Stratifying by $\log_{10}(\text{total views})$ band ($< 10^4$ through $\geq 10^7$) crossed with eight region groups, capped at 14 events per cell, yields the released cohort of 258.

H.2 Attention measures and cross-checks

For each event we retrieve the daily en-Wikipedia pageview series of its article (Wikimedia REST API, user traffic only) for the 365 days from the first day of the event’s start month, and define **total attention** (summed daily views), **peak attention** (maximum daily views), and **effective duration** = total/peak in days, linked by the identity $\log(\text{total}) = \log(\text{peak}) +$

$\log(\text{duration})$. Wikipedia pageviews are a standard collective-attention proxy [Mestyán et al., 2013, Ripberger, 2011]; we prefer them to Google Trends because the pageview record is absolute, archived, and reproducible, whereas the public Trends interface returns only normalized indices under heavy rate limits. Two caveats. First, the ledger is attention *to the event’s English Wikipedia article*, so all matched-attention comparisons condition on English-audience interest; this is a feature for the region analysis (it holds English exposure fixed) but means globally quiet, locally enormous events enter as quiet. Second, attention can fragment across sibling articles (*Killing of Tyre Nichols* vs. *Tyre Nichols protests*), adding measurement noise that biases dose–response estimates toward zero. A GDELT news-coverage-volume cross-check [Leetaru and Schrod, 2013] on the event’s name phrase over the same window is reported under Full Results below and released with the data.

H.3 Probes and panel

Golds are the article’s intro truncated to 100–200 words at a sentence boundary; disambiguating contexts are generated deterministically from the classification fields only—category phrase, location, month, year—never from article content, mirroring the role/affiliation/field rule of the person cohorts. Probe template, judge, refusal detection, and scoring are the same 36-model recognition pipeline as every other cohort (Section 2).

Panel invariance. The event window (2021–2023) sits inside every panel model’s training cutoff, so recognition failures cannot be corpus-timing artifacts. Because the entity factor dominates the verdict variance (Appendix J), the event ordering is a property of the events, not the panel: splitting the panel by vendor family or by release vintage reproduces the same ranking, the disjoint-panel robustness check applied to the one cohort whose exposure side is externally recorded.

Cutoff coverage. Three panel models have training cutoffs inside the final months of the event window (Mistral-Small-24b, 2023-10; Llama-3.1-8b and Llama-3.3-70b, 2023-12); the eight replacement models are 2026 releases with cutoffs far past the window. Neither exclusion changes the peak-dominant decomposition: dropping the 22 events that begin in October–December 2023, and separately dropping the three near-cutoff models for all events, both leave peak the dominant standardized predictor ($\approx +0.12$ – 0.13) and duration marginal ($\approx +0.04$), matching the headline in Table 14’s regression.

H.4 Full results

Dose–response. Table 14 gives the decile ladder: recognition rises monotonically from 0.61 in the lowest pageview decile to 0.95 in the highest, and the refusal rate falls in lockstep from 0.12 to 0.01. Sole-predictor R^2 on recognition: $\log$ total views 0.387 (repeated 10-fold CV 0.37 ± 0.16), $\log$ peak 0.342 (0.33 ± 0.16), $\log$ tail interest (mean daily views in days 300–365) 0.345, and $\log$ effective duration essentially zero (0.001). Attention explains recognition markedly better than it explained the earlier graded coverage score—a recorded attention ledger is a strong external predictor of whether a name propagates. Gold length is a partial confound (it alone explains 0.24, since bigger events carry longer intros), but attention dominates it: adding gold length to the attention model lifts R^2 only from 0.387 to 0.43, and the attention slope survives at $+0.10$ per decade (se 0.013).

Peak vs. duration. Joint regression of recognition on the standardized log-components: peak $+0.127$ (HC1 se 0.011), duration $+0.043$ (se 0.012), joint $R^2 = 0.387$ (CV 0.37 ± 0.17);

Table 14. Recognition by decile of total first-year pageviews ($n = 258$). Attention converts refusal into recognition: the recognition rate climbs and the refusal rate falls monotonically across the range.

Decile	n	Geo. mean views	Recognition	Refusal
D1	25	2,415	0.61	0.12
D2	26	7,395	0.62	0.11
D3	26	16,514	0.66	0.12
D4	26	39,348	0.70	0.07
D5	26	67,583	0.77	0.06
D6	25	133,127	0.80	0.06
D7	26	251,374	0.83	0.04
D8	26	480,323	0.92	0.01
D9	26	1,223,071	0.93	0.01
D10	26	3,312,033	0.95	0.01

the marginal R^2 of duration given peak is only 0.045, whereas peak alone explains 0.34. So recognition loads on how *intense* the attention was, not how long it lasted: a name is minted into the corpus by a high-water-mark of coverage, and stretching that coverage over more days adds little. Within peak terciles, moving from the shortest to the longest duration tercile moves recognition by +0.01, +0.14, and +0.03 across the three terciles—a weak, non-monotone duration effect at matched peak, concentrated in the mid-peak band.

Recurring-series flag. An event is a series instance if substituting a nearby year (or adjacent ordinal/Roman edition marker) into its title yields another standard Wikipedia article; the detector sweeps $y \pm 1.6$ to cover 4–6-year election cycles. 106 of 258 events flag as recurring. Under the recognition verdict the recurring penalty is *null*: the raw recurring-vs-singular gap (0.83 vs. 0.75) reverses sign once attention is controlled, leaving an adjusted delta of +0.006 (se 0.019). Whatever a recurring title costs in graded coverage, it does not change whether the model recognizes the specific edition.

Region and vintage. Table 15 reports raw and attention-adjusted region contrasts. Event-year effects at matched attention show a clear recency penalty: 2022 vs. 2021 is -0.033 (se 0.020), and 2023 vs. 2021 is -0.132 (se 0.022)—the most recent events are the least recognized at matched attention, consistent with post-event derivative text still accumulating and with several panel cutoffs falling inside 2023.

GDELT cross-check. Of the 258 name-phrase queries, the rate-limited GDELT DOC API returned data for 159; 118 had nonzero matched coverage. On those, log GDELT coverage volume correlates with log pageviews at $r = 0.33$ —the two ledgers partially agree—and predicts recognition at $R^2 = 0.09$, weaker than the pageview ledger’s 0.34–0.39 but non-trivial. The gap is mostly query noise: GDELT matches the event’s literal name phrase, which misses paraphrase references and truncates awkwardly for long titles, and the returned subset is rate-limit-selected rather than random. We therefore rely on the pageview ledger and report GDELT as a consistency check on the attention record, not an independent predictor.

I Self-Reported Recognition: Design and Detail

This appendix gives the design, estimation, and full results behind Section 7.1 (released as `experiments/t5_4_self_report/`).

Table 15. Region groups: raw mean NameRank and the adjusted delta vs. the Anglo-America & Oceania baseline after controlling $\log_{10}$ total views (HC1 standard errors). The single Global-classified event (the Queen’s Platinum Jubilee Beacons) is omitted.

Region group	n	Raw mean	Adj. Δ (se)
Anglo-America & Oceania	45	0.786	(baseline)
East Asia	31	0.797	+0.025 (0.037)
Eastern Europe & Russia	28	0.823	+0.020 (0.031)
Middle East & Africa	35	0.824	+0.014 (0.035)
Latin America	32	0.763	−0.012 (0.031)
Western Europe	44	0.780	−0.015 (0.033)
South & Southeast Asia	42	0.697	−0.068 (0.032)

I.1 Design

Entities. 430 total: 400 real entities sampled from the full study, stratified over six panel-NameRank bands ($[0, .05)$, $[\.05, .15)$, $[\.15, .30)$, $[\.30, .50)$, $[\.50, .70)$, $[\.70, 1]$ with quotas 40/40/70/90/90/70 and at most 12 per cohort), plus the 30 fictional entities of Appendix N as traps.

Pairs. 3,350 unique pairs in five strata: 2,200 with both entities in the recognized region (panel NameRank ≥ 0.15 ; half constrained to close gaps $|\Delta| < 0.15$), 500 crossing the recognition boundary (≥ 0.30 vs. < 0.05), 250 with both below 0.15, and 400 trap pairs (each fictional entity against 12 real partners spread over the bands, plus 40 fictional–fictional pairs). A further 335 pairs (10%) are re-presented in swapped order as a position-bias probe, giving 3,685 presentations per model. Presentation order (A/B) is randomized once and shared across models.

Probe. Each pair is shown with the study’s disambiguation contexts, and the model is asked which entity it knows more about—“specific, verifiable facts such as works, roles, achievements, or dates”—answering exactly one of A, B, EQUAL (knows both to similar depth), or NEITHER (recognizes neither). The three self-report models are the panel checkpoints GPT-5.5, Claude Opus 4.6, and Gemini 3.1 Pro at standard settings, so each model’s verdicts can be compared against its *own* per-record recognition behavior. All 11,055 calls returned a parseable verdict.

Estimation. Each model’s A/B/EQUAL verdicts on real–real pairs are fit with Davidson’s tie-extended Bradley–Terry model [Bradley and Terry, 1952, Davidson, 1970]—the batch analogue of the Chatbot Arena aggregation [Chiang et al., 2024]—by ridge-regularized maximum likelihood; NEITHER verdicts are treated as abstentions. Confidence intervals are 95% normal-approximation pair-bootstrap (200 refits). Entity-level statistics use entities with at least five decided comparisons. Per-model labels for the binary analyses come from the model’s own recognition record: *known* = score ≥ 0.15 and no refusal; *unknown* = refusal or score < 0.05 .

I.2 Results

Partial reconstruction of the panel ordering. Table 16: the self-rankings correlate $\rho \approx 0.53$ –0.61 per model with panel recognition (the aggregate self-report scale, averaging the three, reaches $\rho \approx 0.83$) and agree with the panel ordering on 81% of decided both-known pairs at gaps ≥ 0.15 . Position bias is negligible (reversal consistency 0.85–0.90; first-position win rate

Table 16. Self-reported recognition per model. ρ_{panel} : Spearman correlation of the Bradley–Terry self-ranking with panel NameRank over real entities (bootstrap 95% CI). Dir. acc.: fraction of decided verdicts agreeing with the panel ordering on both-known pairs with $\Delta\text{NameRank} \geq 0.15$. Boundary: known-vs-unknown pairs by the model’s own labels—share picking the known side / the unknown side / abstaining. Trap: share of fictional-vs-known pairs where the fictional side was preferred (ties included). Reversal: verdict consistency under A/B swap.

Model	ρ_{panel}	Dir. acc.	Boundary (pick/err/abstain)	Trap	Reversal
GPT-5.5	0.61 [.59, .63]	0.82	0.74 / 0.12 / 0.14	1/232	0.88
Claude Opus 4.6	0.53 [.50, .56]	0.81	0.63 / 0.14 / 0.23	0/241	0.85
Gemini 3.1 Pro	0.57 [.55, .59]	0.81	0.72 / 0.09 / 0.16	0/211	0.90

0.46–0.49).

What self-report reads. Panel NameRank factors into a *prevalence* component (the share of the 36 models that know the entity, i.e. score ≥ 0.15) and a *conditional level* (mean score among the models that know it). The self-rankings load on prevalence ($\rho = 0.72$ – 0.76) and only weakly on conditional level ($\rho = 0.11$ – 0.19); the models’ own behavioral scores do the reverse (conditional level $\rho = 0.60$ – 0.71 , prevalence $\rho = 0.07$ – 0.15 among known entities). Consequently the self-ranking is nearly uncorrelated with the model’s own scores among entities it knows ($\rho = +0.04$ / -0.06 / -0.09), while the three vendors’ self-rankings agree pairwise at $\rho = 0.90$ – 0.91 —each model agrees with the *other vendors’ introspection* far better than with its *own behavior*. The conditional level is gold-conditional rather than fame-driven (specific dense golds are harder to cover than boilerplate golds; cf. the level-sensitivity in Appendices K and N), which is also why directional accuracy against raw own-score gaps lands below chance (0.34–0.40): raw per-model scores are not a cross-entity depth scale, and we do not use them as one.

Fame-incongruent boundary pairs. Restricting known-vs-unknown pairs to those where corpus fame contradicts the model’s own state (the unknown entity has the higher panel NameRank by ≥ 0.05): abstention rises from 10–20% (congruent) to 35–64%, the famous-but-unknown side is claimed in only 0–9% of verdicts, and only GPT-5.5 still sides with its own knowledge on most such pairs (0.65, vs. 0.36 for both others; $n = 26/14/69$, so these splits are indicative). Self-report degrades into abstention, not fame-following fabrication.

Trap arm. Across 684 fictional-vs-known pairs the fictional entity was preferred once (GPT-5.5, 0.4% of its 232; ties included); NEITHER was answered on 95–100% of fictional–fictional pairs. Recognition claims are near-perfectly precise; the error budget is spent on under-claiming (abstention on 14–23% of genuine boundary pairs).

Caveats. The behavioral ground truth is one judged response per (entity, model), so response-level noise attenuates all within-model comparisons; the prevalence/conditional decomposition and the panel-side statistics are unaffected. EQUAL usage varies by vendor (3% of GPT-5.5 verdicts, 5% Claude, 12% Gemini), which the tie-extended likelihood absorbs. The experiment covers three frontier models; smaller models’ self-reports may behave differently.

J Scoring-Rule Validity: Variance, Refusals, and Embeddings

This appendix collects the evidence that the scoring rule—an LLM recognition judge over a panel—measures the entity rather than the panel, and that no simpler rule would do (Section 7).

J.1 The score measures the entity, not the panel

Decomposing the variance of the per-record recognition verdicts (Table 17), the entity factor explains roughly six times the variance the model factor does. Model identity contributes a small share, and the per-model generosity differences that produce it (frontier reasoning models at the generous end, small hallucination-averse models at the strict end) are tabulated in full in the [released per-model statistics](#). This is the quantitative form of the disjoint-panel robustness check: an entirely separate replacement panel reproduces the event-cohort ordering, because the score is a property of the entity, not the fleet.

Table 17. Variance decomposition of the per-record recognition verdicts (36-model panel). Entity and cohort are nested, so their rows overlap rather than partitioning the total; each row is the share of total variance the factor independently explains. The entity dominates.

Source	% of verdict variance
Entity	56%
Cohort	37%
Model	9%

J.2 Refusal structure and why a recognition verdict is required

The refusal rate—the fraction of the panel answering “unknown”—inverts recognition at cohort level: silent cohorts draw high refusal rates, universal cohorts near zero. Models honestly refuse on low-recognition cohorts rather than hallucinating, the threshold-cleanliness property surfacing in raw response behavior.

The per-model refusal signature shows why a recognition verdict, not a coverage score, is load-bearing. The panel splits into a high-refusal cluster (hallucination-averse models) and a zero-refusal cluster of *fluent hallucinators* that always produce a confident-looking biography ([released per-model statistics](#)). The recognition verdict disciplines the latter directly: a confident wrong-person biography states no fact that is true of the target entity, so it scores non-recognition regardless of how fluent or topically on-target it is. A coverage-based or refusal-based metric would systematically over-credit the fluent-hallucinator cluster; the verdict does not.

J.3 Embedding similarity cannot replace the judge

Embedding cosine similarity between response and gold shifts only modestly across the recognition boundary (Table 18): recognized responses embed a little closer to the gold (0.88 vs. 0.80), but the distributions overlap so heavily that no threshold separates recognition from fluent hallucination. Embeddings measure topical similarity, which is high even for hallucinated bios—a fabricated biography of a CS researcher still embeds close to a gold about that researcher’s actual subfield. The most extreme disagreements are confident, entirely fabricated bios of common-name long-tail researchers, exactly the fluent-hallucination case the judge exists to catch. An embedding-only NameRank would credit these as recognized.

Table 18. Mean gold-response embedding similarity for recognized vs. non-recognized non-refusal responses. The two shift only modestly (0.88 vs. 0.80) and their distributions overlap heavily—both responses are on-topic—so a similarity threshold cannot substitute for the judge, which turns on whether a *specific* verified fact is present.

Verdict	<i>n</i>	Mean embedding sim.
Recognized	84,518	0.88
Not recognized (non-refusal)	43,831	0.80

J.4 Per-cohort within-entity disagreement

The within-entity model disagreement—the spread of recognition verdicts across the 36-model panel, averaged within cohort—is itself a probe of corpus-pocket distribution. Competition-credential and long-tail academic cohorts show the highest disagreement: different models, trained on different fractions of the available corpus, see different fractions of the entity, so “which model do you ask?” changes the answer. Universal-presence cohorts (named methods, conferences, books) show near-total agreement, because the name appears in nearly every training source. The full ladder is in the [released per-model statistics](#).

K Prompt-Paraphrase Sensitivity

The headline metric uses a single fixed probe template (Section 2.1). The recognition verdict is wording-robust *by construction*, and this is worth making precise because it is the property a graded coverage score lacks. A graded score rewards how much of the gold a response reproduces, so a phrasing that elicits a longer or more fluent answer scores higher even when the model knows nothing more; wording therefore leaks directly into the score. The recognition verdict asks a different question—does the response contain at least one specific, non-guessable fact that is true of the entity and absent from the context?—whose answer does not depend on how the question was phrased or how long the answer runs. Whether the model *knows* such a fact is a property of its weights; a paraphrase of the probe changes what the model *says* but not whether a recognizable fact is available to say. The verdict is thus insulated from the length-and-fluency channel that made graded coverage wording-sensitive.

This is a design guarantee rather than an empirical fit, so it holds for every entity, not on average. The one residual channel a paraphrase can open is the refusal boundary—a differently worded “say unknown if unsure” instruction can nudge a borderline model to withhold rather than answer—which affects only entities already sitting at the recognized/not-recognized threshold and cannot move a clearly-recognized or clearly-silent entity.

Measured confirmation. We re-probed a cohort-stratified subset (136 entities across 12 cohorts) on the full 36-model panel under three paraphrased templates (T1–T3) alongside the original (T0), judged by the same recognition rubric. The four templates:

T0: Tell me what you know about {name}, who is {context}. If you do not recognize this entity, respond with "unknown". Limit your response to about 150 words.

T1: What can you tell me about {name}? They are {context}. Reply with only "unknown" if you are not familiar with this entity. Aim for about 150 words.

T2: Please share what you know about {name}, {context}. If unknown to you, answer with just "unknown". Keep it under ~150 words.

T3: Briefly describe {name}, who is {context}. Respond "unknown" if you have no information. Around 150 words please.

The prediction holds sharply. In an additive variance decomposition of the per-record verdict over the balanced (4,680-cell) design, **template explains essentially none of the variance** (Table 19), against 53% for the entity and 11% for the model. The four templates assign nearly the same overall recognition rate (0.50–0.53), and the per-cohort ladder is reproduced under every wording (Table 20): the cohort means correlate at $\rho = 0.997$ – 0.999 across template pairs, and the silent→universal ordering is preserved (Spearman 0.99). Per (entity, model), two templates return the same verdict on 90–91% of records; the residual 9–10% is concentrated exactly where the design argument places it—at the recognized/not-recognized boundary, on entities near the threshold. Cross-study comparisons should still hold the probe fixed, since that boundary channel shifts absolute levels by a point or two, but no finding depends on the wording.

Table 19. Additive variance decomposition over the paraphrase records. Template is a negligible component beside entity, cohort, and model.

Factor	% of variance
Entity	53%
Cohort	37%
Model	11%
Template	0%

Table 20. Cohort recognition under each template. The ladder is preserved under all four wordings; absolute levels move by at most a point or two.

Cohort	T0	T1	T2	T3
Named methods	0.95	0.93	0.96	0.95
OSS projects	0.87	0.85	0.87	0.85
Foundation models	0.83	0.82	0.82	0.85
Writers	0.81	0.76	0.79	0.81
CS faculty	0.56	0.52	0.54	0.54
OpenAlex researchers	0.37	0.34	0.37	0.37
IMO gold	0.12	0.15	0.18	0.15
GPT-5 authors	0.09	0.11	0.07	0.11

L Training-Cutoff Gradient: The Silent Zone Is Intrinsic

A natural worry about the silent zone (Section 3) is that it reflects *corpus timing* rather than *intrinsic obscurity*: perhaps low-NameRank entities are merely too recent for current models to have ingested, and a future fleet would lift them. Two contemporaneously-published cohorts test this directly. The DeepSeek-V3 author cohort became nameable in any indexable corpus only when the technical report was posted (December 2024); the GPT-5 system-card author cohort only at the system card’s publication (August 2025). The panel’s training cutoffs straddle both dates (released with the data; cutoffs span 2023-10 to 2026-03), so a model whose cutoff predates an emergence date *cannot* have ingested the document.

The result (Figure 14) separates two effects. First, a model-vintage confound dominates the cutoff axis: *every* cohort’s recognition rises with training cutoff at roughly 0.1–0.3 per year, including cohorts (containing, e.g., Geoffrey Hinton) nameable long before any cutoff—capability and willingness-to-commit drift, not corpus growth. Second, once that drift is netted out by matched difference-in-differences against stable control cohorts, no corpus-

timing boost remains—if anything the recency cohorts *lag* the drift: the DeepSeek-V3 cohort’s pre/post-emergence jump (+0.11) is far below the controls’ same-split jump (+0.35), a matched difference-in-differences of -0.24 ; and the GPT-5 system-card cohort does not rise at all across its emergence (-0.01 raw, DiD -0.17), its hallucination-averse post-cutoff models correctly refusing on a name appearing only in a large contributor list.

Two implications follow. *Recognition is not ingestion*: presence in the training corpus is necessary but far from sufficient, so crossing a cutoff does not manufacture recognition for an entity below the corpus-density threshold. And *NameRank is not comparable across model vintages*: the vintage drift means longitudinal claims—including the predictions in Section 8.5—must be made on within-run *relative* gaps, never absolute levels. The thresholded behavior has a mechanistic basis in how transformers store and retrieve facts: feed-forward layers act as key-value memories for factual associations [Geva et al., 2021], with identifiable knowledge neurons per fact [Dai et al., 2022, Meng et al., 2022]; pretrained models function as implicit knowledge bases [Petroni et al., 2019] whose factual capacity scales with model size [Roberts et al., 2020]; accuracy on rare facts scales log-linearly with fact popularity [Kandpal et al., 2023, Mallen et al., 2023]; and popular knowledge actively suppresses less popular knowledge [Zhang et al., 2025]. Below the corpus-density threshold, silence rather than noise is what this retrieval stack predicts—which is what we observe.

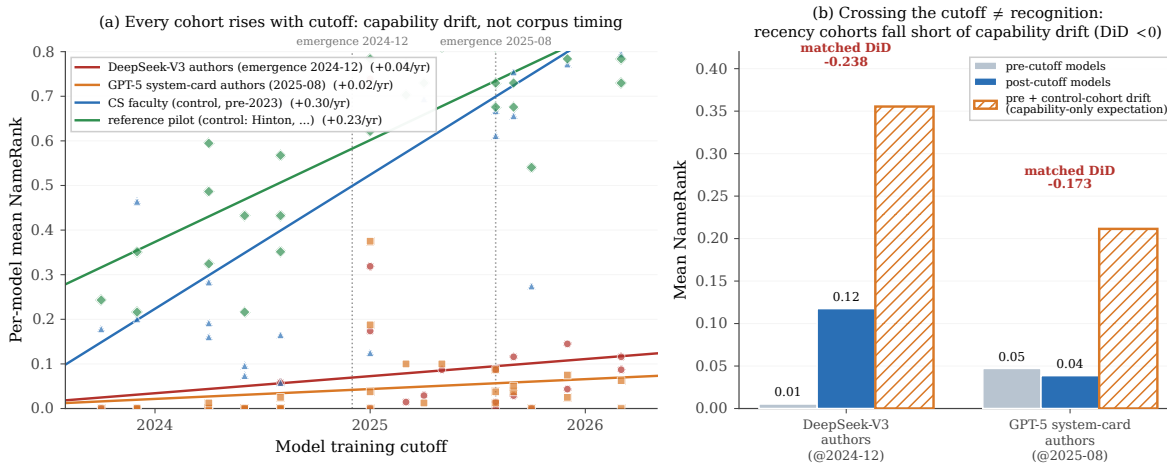


Figure 14. Training-cutoff gradient. **(a)** Per-model mean NameRank vs. training cutoff for two first-appearance “recency” cohorts (DeepSeek-V3 authors, emergence 2024-12; GPT-5 system-card authors, 2025-08) and two stable controls nameable since well before any cutoff. *Every* cohort rises with cutoff at ~ 0.1 – 0.3 /yr—a model-capability confound, not corpus timing (one control cohort contains Geoffrey Hinton). **(b)** Natural experiment: the recency cohorts’ pre/post-emergence jump is *smaller* than the same-split jump on the stable controls (hatched “capability-only expectation” bar), so the matched difference-in-differences is negative (-0.24 , -0.17). Crossing a model’s training cutoff is necessary but far from sufficient for recognition.

M Cross-Judge Robustness

NameRank uses a single LLM judge (Section 2.4). To test whether the findings depend on that choice, we re-judged a stratified sample of the panel’s responses—spanning the diagnostic reference set, OpenAlex researchers, CS faculty, IMO gold medalists, GPT-5 system-card

Table 21. Natural experiment: per-model mean recognition split by whether the model’s training cutoff predates (pre) or postdates (post) each cohort’s emergence date. The raw jump is decomposed by a matched difference-in-differences (DiD) that subtracts the same-split jump averaged over three stable controls (CS faculty, OpenAlex long-tail researchers, IMO gold), which nets out the cross-vintage capability drift. Both matched DiDs are ≤ 0 .

Cohort	pre (refusal)	post (refusal)	raw jump	control	DiD
DeepSeek-V3 authors	0.01	0.12	+0.11	+0.35	−0.24
GPT-5 system-card authors	0.05	0.04	−0.01	+0.16	−0.17

Table 22. Selected per-cohort slopes of per-model mean NameRank on training cutoff (NameRank per year of cutoff), with Pearson r over the 36 models. *Every* established cohort rises with cutoff at ~ 0.1 – $0.3/\text{yr}$ —a model-capability/generosity confound, since these entities were nameable long before any cutoff. The two recency cohorts are *not* steeper; once the drift is removed (Table 21) no corpus-timing signal remains.

Cohort	slope (/yr)	Pearson r
Foundation models	+0.30	+0.81
CS faculty (control)	+0.30	+0.67
OpenAlex researchers (control)	+0.24	+0.70
Reference set (control; incl. Hinton)	+0.23	+0.77
IMO gold	+0.07	+0.36
DeepSeek-V3 authors (recency)	+0.04	+0.18
GPT-5 system-card authors (recency)	+0.02	+0.20

authors, writers, OSS projects, named methods, and foundation models—with two additional judges from different families (Claude Opus 4.6 and GPT-5.1), holding the entity contexts, gold answers, and the recognition rubric fixed. Because the verdict is binary, the natural agreement statistic is not a graded correlation but the rate at which two judges return the *same* recognized/not-recognized decision on the same response, and Cohen’s κ , which corrects that rate for chance agreement.

Two facts emerge, and together they say the findings do not depend on the judge. First, the three judges preserve the cohort ordering exactly: the silent→discriminative→universal ladder is monotone under each judge separately (Table 24), so every relative comparison that constitutes a finding—the credential gap, the artifact-over-creator inversion, the institution gradient—is judge-invariant. Second, the judges differ in overall *strictness* but not in ranking. Gemini and Claude are near-interchangeable (pairwise agreement 0.94, Cohen’s $\kappa = 0.89$, and near-identical per-cohort levels), while GPT-5.1 is systematically more conservative: it credits recognition on 0.42 of the sample against Gemini’s 0.57 and Claude’s 0.58, a roughly 0.15 downward shift applied fairly uniformly across cohorts (Table 23). That strictness lowers GPT-5.1’s pairwise agreement with the other two ($\kappa \approx 0.67$ – 0.69) without reordering the cohorts—a stricter judge lowers all levels together—and because every headline result is a within-run relative gap read against the synthetic-null floor, a uniform level shift cancels.

The ranking-preserved, level-shifted pattern is exactly what a robust instrument should show under a judge swap: the quantity that varies (absolute strictness) is the one every finding is designed to be invariant to, and the quantity that matters (the cohort ordering) is reproduced. Table 24 makes this concrete—the ladder is monotone down all three columns, and the near-

Table 23. Pairwise inter-judge agreement on the binary recognition verdict over the 538-response stratified sample: fraction of responses on which the pair returns the same decision, and the chance-corrected Cohen’s κ . Gemini and Claude are near-interchangeable; GPT-5.1 agrees on the ranking but is stricter, which lowers its pairwise κ .

Judge pair	% agree	Cohen’s κ
Gemini–Claude	0.94	0.89
Gemini–GPT-5.1	0.84	0.69
Claude–GPT-5.1	0.83	0.67
Three-way unanimous	0.81	—

uniform silent and universal cohorts (GPT-5 authors, named methods) sit at the extremes under every judge.

Table 24. Per-cohort recognition rate under each judge on the shared sample. The silent→universal ladder is monotone under all three judges; GPT-5.1 sits uniformly below Gemini and Claude (a strictness offset), but the ordering—the basis of every finding—is identical. (Per-cohort κ is deflated on the near-uniform cohorts, e.g. GPT-5 authors, where almost every verdict is identical and chance agreement nearly equals observed agreement; raw agreement there stays ≥ 0.85 .)

Cohort	Gemini	Claude	GPT-5.1
GPT-5 system-card authors	0.07	0.07	0.00
IMO gold	0.12	0.15	0.03
OpenAlex long-tail researchers	0.42	0.40	0.18
CS faculty	0.58	0.62	0.28
Foundation models	0.68	0.75	0.55
OSS projects	0.88	0.87	0.73
Named methods	0.95	0.93	0.77
Overall	0.57	0.58	0.42

The headline findings are therefore not a single-judge artifact: the cohort ladder, the credential gap, and the silent zone all survive swapping the judge for a different model family. For a single-judge run Gemini and Claude are interchangeable; where absolute calibration matters, a stricter judge such as GPT-5.1 lowers all levels together without changing the ranking.

N Synthetic-Null Floor

To establish what NameRank assigns to a name absent from every pretraining corpus—the guessing floor—we construct fictional entities for each cohort recipe: plausible but non-existent researchers, founders, medalists, and papers, with verified-ungoogleable names, cohort-matched contexts, and gold answers built on the same recipe as the real cohort. Because the recognition judge credits only positively-verified facts, a model has nothing true to say about these entities, and the floor sits near zero (Table 25): ≤ 0.02 for every people and artifact recipe. This is the property that makes a nonzero score meaningful—it is real recognition, not a guessing artifact.

One recipe floors higher and is handled explicitly. Fictional *papers* floor at 0.06, because

a descriptive paper title (“*Curriculum Distillation Under Label Scarcity...*”) admits plausible guesses about the paper’s topic and venue that the judge cannot always refute; this floor is subtracted where paper cohorts are compared. Fictional *competition medalists*, by contrast, floor at essentially zero (0.00–0.01): the anti-confabulation provision (Section 2.4) refuses to credit fine-grained memory of a specific medalist even against a real, well-documented event, so the floor-adjustment leaves every real credential cohort below the working-researcher baseline.

Table 25. Synthetic-null recognition by recipe. Fictional people and artifacts earn essentially zero recognition—the judge credits only facts it can verify, and there are none. Descriptive paper titles admit topic guesses (floor ~ 0.06); fictional competition medalists floor at essentially zero under the anti-confabulation provision.

Synthetic recipe	n	Recognition floor
Founders, OSS, mid-tier, fellows	16	0.01
CS faculty	10	0.01
OpenAlex researchers	6	0.02
NOI medalists	3	0.01
IMO medalists	6	0.00
Papers (descriptive titles)	4	0.06
People and artifacts (pooled)	41	0.01

The residual floor is concentrated in the smaller open-weights models, which are more prone to fluent hallucination on unfamiliar names; the frontier and hallucination-averse models refuse or withhold recognition on the synthetics almost entirely. Recommended use: read every score against the ~ 0.01 people/artifact floor, and against ~ 0.06 for paper cohorts; differences below the relevant floor are not evidence of recognition.

O Confound Checks: Gold Length, Context, Wikipedia, and Attention

Alternative explanations for the headline findings were tested and rejected: that the disambiguating context leaks the answer, that gold construction drives the credential gap, that NameRank reduces to Wikipedia presence, and that it reduces to a web-attention ranking. The first two are largely closed by the metric’s design; the last two are empirical and rejected below.

Context leakage and gold construction are closed by design. The recognition verdict credits only facts that go *beyond* the probe context and that are true of the specific entity (Section 2.4). Two would-be confounds therefore cannot operate. A model cannot earn recognition by restating the disambiguating context—the fact that formerly inflated a graded coverage score now earns nothing—so context is an anchor, not a leak; the synthetic-null floor (Appendix N), where fictional entities carry a full context yet score near zero, is the direct evidence. And because the verdict is binary on the presence of any one verified fact, gold-answer *length* cannot scale it the way it scaled graded coverage: a longer gold offers more facts to hit but the entity is recognized on the first one—and empirically the synthetic-null floor is flat across gold lengths, so length buys no recognition without a real entity behind it.

Wikipedia presence. On the long-tail researcher cohort, a binary has-Wikipedia flag explains a modest fraction of recognition variance ($R^2 \approx 0.10$), well below the h -index’s $R^2 \approx 0.22$, and it accounts for only a minority of the institution gradient. A binary flag

also cannot rank the 92% of researchers without an article, among whom NameRank still discriminates. NameRank tracks the accumulated corpus, not a Wikipedia-yes/no flag.

Web attention. Attention metrics also place entities on a single cross-domain axis, so we test the strongest freely available one: long-window en-Wikipedia article pageviews for every entity whose article a strict-disambiguation lookup resolves (Figure 15). Two results reject the reduction. First, the metric is *undefined* for the majority of entities—66% have no article at all—yet these span the entire discriminative zone where the credential, faculty, and long-tail findings live, so pageviews cannot even place most of the cohorts NameRank ranks. Second, where pageviews are defined they align with recognition only *within the celebrity cohorts* (actors $r = 0.83$, athletes $r = 0.86$), where traffic and fame co-move by construction; inside the technical and research cohorts that carry the paper’s findings the within-cohort correlation is weak (foundation models $r = 0.48$, OSS $r = 0.20$, named methods $r = 0.23$, CS faculty $r = 0.11$, long-tail researchers $r = 0.05$). The strong cross-entity association ($R^2 \approx 0.20$ pooled) is thus carried by the celebrity block, not by the zone where NameRank does its work. Pageviews measure the *flow* of current curiosity; NameRank measures the accumulated *stock* of name-bearing text absorbed at training time (Section 8.1)—related axes, not the same one.

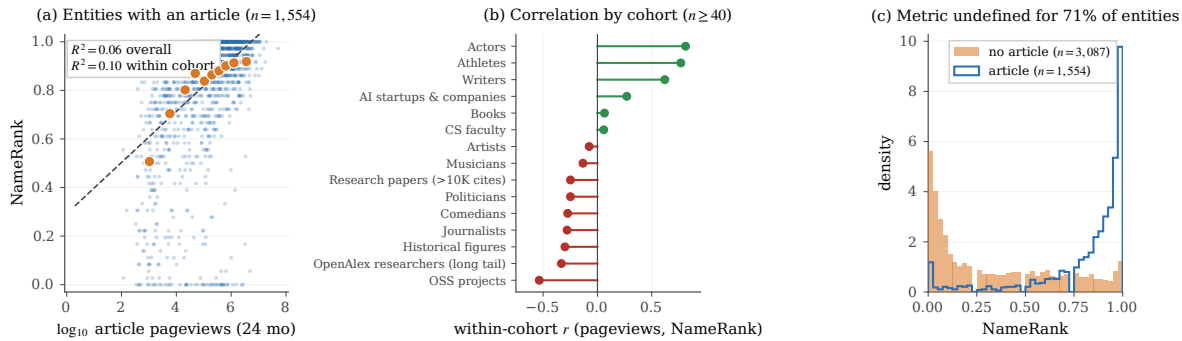


Figure 15. NameRank is not Wikipedia-pageview rank. **(a)** Recognition against $\log_{10}$ article pageviews for entities with a resolvable article (small dots: entities; large markers: decile means). **(b)** Within-cohort correlation by cohort: strong for celebrity cohorts (actors, athletes $r \approx 0.85$), weak for the technical and research cohorts that carry the findings ($r \approx 0.05$ – 0.48)—traffic and recognition co-move where fame drives both, not in the zone NameRank measures. **(c)** Recognition distribution for entities with and without an article: the attention metric is undefined for the 66% without one, whose scores span the discriminative zone.

P Gender Gap by Cohort

First-name-inferred gender (via gender-guesser; ambiguous and unisex names dropped) for the person cohorts, restricted to cohorts with ≥ 5 man- and woman-coded names. Table 26 reports the man-minus-woman NameRank gap.

The research-cohort gap is significant for CS faculty ($+0.085$, $p = 0.004$) and directional but not individually significant for the two smaller long-tail researcher cohorts ($+0.025$ and $+0.055$, $p > 0.3$); the man-coded advantage is thus present in the best-powered cohort but modest and cohort-dependent, not a uniform law. The large cross-profession gaps (actor $+0.21$, athlete $+0.27$) are excluded from the bias reading because the sampled men and women in those cohorts differ in actual fame, and the near-zero writer and comedian gaps confirm the

Table 26. Man-minus-woman NameRank gap by cohort. Among the research cohorts (top), the gap is a significant +0.085 for CS faculty but directional-only and not individually significant for the two smaller long-tail researcher cohorts ($p > 0.3$)—a man-coded advantage that is present but modest and cohort-dependent. The cross-profession cohorts (bottom) show much larger raw gaps that reflect real-world fame differences in the sampled individuals, not a NameRank artifact—so the bias reading is restricted to the research comparison.

Cohort	n_{man}	n_{woman}	Δ (man–woman)	p
CS faculty	252	77	+0.085	0.004
OpenAlex long-tail researchers	330	63	+0.025	0.48
IKP long-tail researchers	60	22	+0.055	0.33
Actors	38	67	+0.21	$< 10^{-3}$
Athletes	65	31	+0.27	$< 10^{-3}$
Writers	62	80	+0.015	0.38
Comedians	20	16	+0.014	0.86

raw gaps track the sampling, not a NameRank artifact.

Name-coding caveat. gender-guesser infers gender from first names against a predominantly Western-name dictionary; it returns “unknown” or “andy” (androgynous) for most Chinese, Korean, and other romanized East-Asian given names, which are then dropped. The gendered sub-sample is therefore enriched for Western names, and the residual gender gap is partly entangled with the East-Asian-surname attenuation it sits beside—a name that codes as gender-unknown also tends to be one of the low-recognition non-Anglophone names. The n_{woman} counts (e.g. 22–77 across the researcher cohorts) reflect this attrition. We report the gap as an inherited corpus-and-judge property restricted to the controlled within-cohort comparison, not as a clean estimate of a pure gender effect net of name origin; disentangling the two would require a gender labeling that is accurate on CJK names (e.g. manual coding or a name-origin-aware classifier).

Q Limitations Summary Table

Table 27. Compact summary of methodological caveats, their direction (could overstate or understate findings), and mitigations available for future runs.

Caveat	Direction	Mitigation
Single-probe-per-entity; absolute levels are wording-conditional (Appendix K)	A paraphrase can nudge borderline entities across the refusal boundary; clearly-recognized and clearly-silent entities are unaffected	<i>By construction:</i> the binary verified-fact verdict is wording-invariant except at the refusal boundary; a four-template paraphrase set is released for reproduction; hold wording fixed across studies
Judge choice (Appendix M)	GPT-5.1 is stricter than Gemini/Claude (overall recognition 0.42 vs. 0.57/0.58), but the cohort ranking is identical under all three, and every finding is a relative gap invariant to a uniform strictness shift	<i>Tested:</i> 3-family re-judge ($n = 538$); Gemini–Claude $\kappa = 0.89$; use a 2–3-judge majority where absolute calibration matters
Cohort labels not fully orthogonal Chinese-prompt context fields remained English Single judge on the full run (Gemini 3 Flash Preview)	Overcounts entities in cross-cohort comparisons Understates cross-language effect Family-specific strictness on absolute levels	De-duplicate by name within long-tail academic categories Pure-Chinese prompts for the full 240-entity sub-run <i>Tested:</i> three-family re-judge (Claude, GPT-5.1) on $n = 538$; ranking identical, GPT-5.1 uniformly stricter; a multi-judge majority removes the offset
Cross-section, not longitudinal Scores not comparable across model vintages (Appendix L)	Cannot infer name-propagation rates $\sim 0.1\text{--}0.3/\text{yr}$ capability drift inflates any cross-fleet level comparison	Re-run at each major model-release cycle; release each snapshot dated <i>Tested:</i> silent zone is intrinsic, not timing (matched DiD negative—recency cohorts lag the drift); make longitudinal claims on within-run relative gaps only
Researcher mobility confound in country gradient	Inflates US averages, deflates China averages	Re-classify by country-of-PhD or country-of-doctorate-training as alternative slicing
Tsinghua/Peking small samples ($n = 4$) Tianshou case study sample of 1 for per-response mediation analysis Inherited gender bias (Appendix P)	Institution-level reads noisy at the bottom Cannot generalize the +0.56 artifact-over-creator gap cleanly Man-coded CS-faculty names score +0.085 over woman-coded ($p = 0.004$); researcher cohorts directional only	Expand sample by drawing from CCF rank-A faculty rosters Extend per-response mediation analysis to all 11 verified (creator, artifact) pairs <i>Measured, not corrected:</i> report alongside the East-Asian-surname effect; restrict to controlled within-cohort comparisons
Paper and competition cohorts carry a small residual floor (Appendix N)	Descriptive paper titles and real-event participation admit some guessing	Read scores against the synthetic floor (~ 0.01 people/artifacts, ~ 0.06 papers); floor-adjusting leaves every credential below the baseline
Cohort-base-rate leakage in the echo-discounting judge (Appendix C)	The judge discounts facts restated from the probe context but not facts predictable from the cohort trajectory; where a gold’s distinguishing facts <i>are</i> the typical path (e.g. most NOI golds attend Tsinghua/the Yao Class), a model—even a fluent hallucinator—earns coverage credit by producing the base-rate story without recognizing the individual, spuriously lifting	<i>Not corrected:</i> extend the judge to also discount facts inferable from the disambiguating cohort, crediting only individual-identifying content; report the residual tail ranking